

Spaghet Boi's Handbook of Handbuilding

December 1, 2024

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1 Tile Efficiency

Tile efficiency is a set of policies and techniques that allows us to reach an optimal point when taking speed, value, win rate, and many other factors into consideration. There is also the concept of *pure* tile efficiency which concerns of only the speed of reaching *tenpai*, one tile away from completing our hand.

In most rulesets such as M. League ruleset where there are yakus such as dora, akadora, ippatsu, and uradora to inflate the expected value of riichi, the best course of action is often to become the first player to call riichi, and the speed and opportunity of reaching tenpai depend on how wide the acceptance of our hand is. Therefore, pure efficiency is very often prioritized and preached. Indeed, mahjong is not a math contest where the player who can calculate the most acceptance automatically wins the game; there are many other factors that would require us to sacrifice efficiency for a better overall outcome, and we will see how those factors play in a bigger picture of hand building.

The following text assumes that the reader has a basic understanding of mahjong rules such as suits, tiles, win conditions, call mechanisms, and riichi. There are, however, many terms that I cannot find a proper translation in English, so I will use Japanese terms as is. Some of them are listed in Table 1. Many other terms such as *ryanmenkanchan* and *agaritoppu* will be introduced later in the text.

1.1 Complex Taatsu

Complex taatsus are shapes involving simple taatsus (penchans, kanchans, and ryanmens) and an additional tile. Table 2 lists complex taatsus that uses three tiles.

Indeed, there are other kinds of complex shapes such as , and so on. Shapes with mentsus involved have more tiles and thus less frequent than shapes with less tiles, but their complexity can still easily confuse us. In Section 1.1, we will deal with complex shapes that has no mentsu inside. Shapes with mentsus will be covered later.

Term	Definition
Ryanmen	The shape consisting of two neighboring number tiles such as  (23p)
Kanchan	The shape consisting of tiles whose numbers have difference of 2 such as  (24p)
Penchan	The shape consisting of number tiles of 1 or 9 and a neighboring tile such as  (12p)
Toitsu	The shape consisting of two identical copies such as  (11p)
Taatsu	The set of ryanmen, kanchan, and penchan
Shuntsu	The shape consisting of three consecutive number tiles such as  (123p)
Koutsu	The shape consisting of three identical copies such as  (111p)
Minko	A koutsu resulting from pon call and thus open to all players
Anko	A koutsu in our hand invisible to other players
Mentsu	The set of koutsu and shuntsu, completed shapes
Tenpai	One tile away from completing our hand
1-away	One tile away from tenpai, also called <i>iishanten</i>
2-away	Two tiles away from tenpai, also called <i>ryanshanten</i>
Shanpon	A wait pattern where we have two pairs in our hand

Table 1: Terminology

Name	Shape	Acceptance
Penchan-toitsu		 6 tiles
Kanchan-toitsu		 6 tiles
Ryankanchan		 8 tiles
Ryanmen-toitsu		 10 tiles

Table 2: Types of complex taatsus using three tiles

1.1.1 Complex Taatsus with Pairs

Since complex taatsus from Table 2 (except for ryankanchan) contain a pair, they must serve as head when there is no other head candidate in our hand. On the other hand, when we have more than two complex taatsus containing pairs, the pairs will impair our acceptance, so in most cases we opt to break those pairs and reduce them to single taatsus. In order to increase the probability of reaching good waits, as a rule of thumb, we drop one tile from the strongest complex taatsu (ネマタ, 2014m,n). Table 3 lists our actions based on the number of pairs, complex taatsus (excluding ryankanchans), and ryanmen-toitsus in our hand.

Pair	Complex taatsu	Ryanmen-toitsu	Action
2	2		Reduce the weakest complex taatsu to a pair
3	2	0	Break the pair in the inner kanchan-toitsu Reduce outer penchan-toitsu to a pair
3	2	1	Reduce the ryanmen-toitsu to a ryanmen
3	3	0	Reduce the weakest complex taatsu to a pair
3	3	1	Reduce the ryanmen-toitsu to a ryanmen
3	3	2	Reduce the weakest complex taatsu to pair

Table 3: Action based on the number of pairs, complex taatsus, and ryanmen-toitsus.

Note that Table 3 is just a rule of thumb. Depending on many other factors such as the existence of yakuhai pairs or table situations, we may take alternatives.

Two Pairs

We must drop a tile from either or from (1) since we will not be touching completed groups and simple ryanmens. To keep two pairs within our hand, our choices narrow down to and . Their acceptances are listed in Table 4, which suggests that dropping has 4 more tiles than dropping in terms of acceptance. This is a case of reducing the weaker complex taatsu to a pair and keeping the stronger complex taatsu, and this particular shape of 1-away is called *perfect 1-away* or *kanzen iishanten* in Japanese.

(1)

Drop	
Acceptance	20 tiles

Table 4: Acceptance of (1).

(2) has which makes dropping and reducing it to ryanmen attractive. But by doing so, we lost flexibility in manzu shape and we must make either or our head, thus reducing our acceptance. If we see (2) as choosing between ¹ and , it becomes clear that dropping here gives us the widest acceptance. Now, dropping also gives the same acceptance as , but considering the case when we draw and wait on , dropping here makes the outside tile more likely to come out. Hence drop (渋川難波, 2024b). Table 5 lists our acceptance.

(2)

East-1, East seat, turn 6, dora
Drop
Acceptance

Table 5: Acceptance of (2).

If our choices are between two ryanmen-toitsus, reduce the one on the inside into a pair to make sure that when we reach tenpai our waits are outside, which has higher win rate. Therefore, between and from (3), we drop and keep the possibility of wait (お知らせ, 2020, pp. 208–209).

(3)

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¹Strictly speaking, can be treated as either + or + , depending on which mentsu we extract from it. Either way, manzu shape is made of a ryanmen-toitsu and a mentsu; it does not change the fact that we are comparing between a ryanmen-toitsu and a penchan-toitsu.

(4) also has two ryanmen-toitsus  and , and one is connected to a completed mentsu . Upon seeing this complex shape, one may be reluctant to touch complex souzu shape and tempted to drop  to fix  ryanmen or drop . The  drop forces us to treat  as our head and renders  surplus, thus having less acceptance than  drop, while  drop leads us to final wait on either  or . Our previous conclusion applies here as well: drop  since the ryanmen-toitsu  is inside (渋川難波, 2024b). Table 6 lists our acceptance.

East-1, East seat, turn 6, dora 

(4)               

(6)

For complex taatsus with few ryanmen-improvement draws, if the remaining pair is in the middle of a suit, we can expect its ryanmen improvement. We need to reduce a complex taatsu into a pair for that outcome, and it then becomes optimal to fix the weakest complex taatsu into a pair. (7) has that, when turned into a penchan (simple taatsu), has no ryanmen improvement, compared to which produces a ryanmen when we draw so we drop and make it a pair (ネマタ, 2014m).

(7)

(8) has a more complex shape that can be treated as either + + or + which leads to its wide acceptance of . We will lose some of its acceptance if we drop any tile from this shape, so instead we reduce to a ryanmen, since technically we still have three pairs in our hand. Table 8 lists the acceptance after different drops. This leads to the rule of thumb that we keep the more complex shape with more acceptance intact (G・ウザク, 2016, p. 24).

(8) East-1, North seat, turn 7, dora

Drop																		
Acceptance	24 tiles																	

七萬 八萬 九萬

South-3, East seat, turn 5, dora 



If there are three complex taatsus, all of which contain no ryanmen, then it is optimal to reduce the one with least ryanmen improvement into a pair, so that when the remaining complex taatsus becomes a mentsu, there will be no surplus tiles. (14) is the hand derived from (13) by dropping and completing . (15) is derived from (13) by dropping instead and completing . We can see, in Table 9, that dropping loses acceptance and we have 2 tiles of acceptance less than dropping , but once becomes a mentsu, (14) has 4 tiles of acceptance more than (15). We therefore prioritize the width of 1-away acceptance and sacrifice a little bit at 2-away. Drop (ネマタ, 2014m).

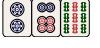
(13) 

(14) 

(15) 

[illegible]

7

drop candidates  from (16), but there will be huge differences when we draw any of the acceptances, as shown in Table 10. If we drop , completing the remaining two complex taatsus  and  will leave us only 3 kinds 12 tiles of acceptance. Likewise for  drop. On the other hand, if we drop , all effective draws will guarantee us 16 tiles of acceptance. One may be tempted by the  draw after dropping  which gives us perfect 1-away, but also keep in mind that it is more likely to draw ourselves into 3 kinds 12 tiles of acceptance.

(16) 

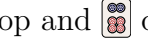
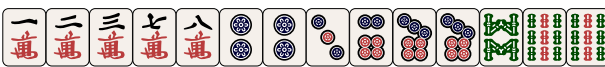
Drop	Next draw	Max acceptance
		drop  →  16 tiles
		drop  →  16 tiles
		drop  →  16 tiles
		drop  →  16 tiles
		drop  →  16 tiles
		drop  →  16 tiles
		drop  →  16 tiles
		drop  →  16 tiles
		drop  →  12 tiles
		drop  →  12 tiles
		drop  →  20 tiles
		drop  →  12 tiles
		drop  →  12 tiles

Table 10: Differences between  drop and  drop from (16)

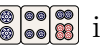
Since it is difficult to calculate all the acceptance of all possible effective draws after dropping each possible tiles, we can just follow the principle to reduce the strongest complex taatsus to a ryanmen.

Three Pairs, Three Complex Taatsus, Two Ryanmen-Toitsus

If there are two ryanmen-toitsus, then we reduce the remaining complex taatsus to a pair. After that, since we have three pairs in our hand, we will have 2 tiles of acceptance less than reducing the other taatsus to a ryanmen. Considering the acceptance when we reach 1-away, however, it becomes clear that this sacrifice of 2 tiles at 2-away is worth. If we choose to fix any ryanmen in (17), completing any complex taatsus leaves us 1 surplus tile. On the contrary, by making the weakest complex taatsus our head, we are guaranteed perfect 1-away no matter what we draw, as shown in Table 11. Drop , even though it may be a little bit counterintuitive because we have been hearing that three pairs are the weakest.

(17) 

1.1.2 Ryankanchan and Its Derived Shapes

The shape  is called *ryankanchan* in Japanese, sometimes *ryankan* for short. The shape contains two kanchans  and , and *ryan* in Japanese means “two”, hence the name. It

Drop	Next draw	Max acceptance
		drop → 20 tiles
		drop → 16 tiles
		drop → 16 tiles
		drop → 16 tiles
		drop → 20 tiles
		drop → 20 tiles
		drop → 20 tiles
		drop → 20 tiles
		drop → 20 tiles
		drop → 20 tiles
		drop → 20 tiles
		drop → 20 tiles
		drop → 20 tiles

Table 11: Differences between dropping and from (17)

takes up the space of 3 tiles and accepts at most 8 tiles, and we are left to chose any of its two kanchans when other parts of our hands get filled up, so ryankanchan is weaker than ryanmen in terms of space efficiency and final waits.

Comparing Ryankanchans

Even between different kinds of ryankanchans, there are differences other than the tiles used. Table 12 lists three kinds of ryankanchan , , and . Note that we omit and because they are identical to and due to symmetry of mahjong tiles. We can see that has the most improvement draws, and dropping from makes our suji wait better than non-suji kanchan wait. and the ryanmen developed from it are all tanyao tiles, so it is a shape we would like to keep for tanyao.

Shape	Ryanmen draw	Comment
		Dropping makes wait suji
		Dropping still keeps the acceptance of

Table 12: Comparison between ryankanchans

Comparing Ryankanchan with Other Taatsus

In the case where there are no floating tiles to drop and we must choose between ryankanchan and other taatsus, when we choose to drop reduce the ryankanchan to a simple kanchan, we lose 4 tiles of acceptance. Therefore, to destroy ryankanchan or not depends on the trade-off of the loss of 4 tiles of acceptance:

1. If destroying ryankanchan still gives us the maximal acceptance, it is the right choice
2. If there is no difference in terms of acceptance,
 - (a) When pinfu is feasible, keep ryankanchan

(b) When pinfu is impossible, drop ryankanchan and set up suji trap

We must drop a tile from either or (18). Notice that the complex pinzu shape has an acceptance of 12 tiles. If we drop 7 tiles; if we drop

 improves our souzu shape, and any draw of manzu tile that gives us head leaves us ryanmen tenpai. The problem becomes which of  to drop. Considering the acceptance of  and the improvement draw , drop  (G・ウザク, 2019, p. 84).

(21)

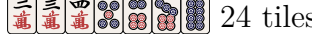
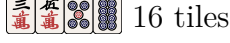

Drop   
Acceptance  24 tiles  24 tiles  16 tiles

Table 15: Acceptance of (21).

Since (22) has two ryanmens with overlapping acceptance, dropping ryanmens and dropping any of  from  will not lead to any difference in terms of the number of acceptance. However, considering the shape of our final waits, we can't touch ryanmens. We must keep  pair in case we draw any of  which gives us perfect 1-away. We also cannot overlook the possibility of 678 sanshoku, so we are incentivized to keep . In this regard,  becomes the most unwanted tile. Drop  (G・ウザク, 2016, p. 14).

(22)

East-1, East seat, turn 5, dora 


Drop   
Acceptance  24 tiles  24 tiles  24 tiles

Table 16: Acceptance of (22).

When the 5 of 135 Ryankanchan is Akadora

The draws that are related to the akadora of  ryankanchan are , which makes  seemingly useless. That said, when we draw ,  becomes floating and  draws make  useful. Besides, when we draw another , it produces a head and a kanchan; drawing  further develops it into a ryanmen. Therefore, we would like to keep it intact, unless we have enough blocks to aim for tenpai (ゆうせー, 2024b).

For example, in (23) since we only have  +  +  +  four blocks, we must drop  and aim for draws that give us another block. If we draw , as in (24), then we can drop  and expect blocks around floating  and .

(23)

East-1, turn 1, dora 



(24)

East-1, turn 2, dora 



advance to make our suji wait  more likely to come out⁷. When our hand is callable, the  pair becomes more valuable.

Drop			
Result shape	  	  	  
Pinfu draws	 4 tiles	 8 tiles	 4 tiles
Ryanmen improvement draws	  12 tiles	 4 tiles	  8 tiles
Final waits	 shanpon	 kanchan	 kanchan or  shanpon

Table 18: Comparing drops from the shape    .

(32) has pinfu chance if we drop  and draw . Meanwhile, (33) cannot have pinfu no matter what we drop; in this case, drop  and make  more likely to come out. (34) has no pinfu either, and it has  anko, which means that, after dropping , calling pon on both  will give us tenpai (ゆうせい, 2021d).

(32) East-1, East seat, turn 5, dora 
                         

(33) East-1, East seat, turn 5, dora 
                         

(34) East-1, East seat, turn 5, dora 
                         

Horizontally Developed Ryankanchans such as 2357

Drop the kanchan part; generally, drop 5 first; when we have no head, drop 7 (ゆうせい, 2022a). Even if tanyao is within reach, keeping ryankanchan takes up the space which we could use for safe tiles instead. It is also possible that our ryankanchan ends up a kanchan bad wait, which leaves us no reason to not destroy it when we have a ryanmen.

When we add  to                         

(35) East-1, turn 3, dora 

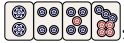

In addition to ryanmen improvement draws, considering final waits, the kanchan part of  is inferior to outside kanchans⁹. In this regard, drop  and keep  in (36).

(36) East-1, turn 3, dora 


When we drop  from , the  shape produces a head if we draw any of . Drawing  gives us a yonrenkei shape¹⁰, which also produces a head easily. Since (37) has no head, drop .

(37) East-1, turn 3, dora 


Ryankanchans with Following Ryanmens such as 34468

The  shape produces a mentsu if we draw any of , as shown in Table 19, and therefore is a strong shape (クリアレイン, 2024; ゆうせー, 2022c).

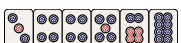
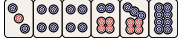
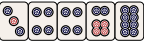
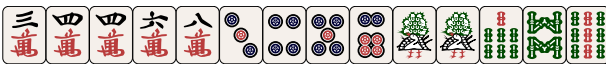
Draw	Result shape	Composition
		Shuntsu  + ryankanchan 
		Anko  + kanchan  or ryanmen-toitsu  + ryankanchan 
		Ryanmenkanchan (see 1.3)
		Ryanmen-toitsu  + shuntsu 

Table 19: Draws that produce a mentsu out of  shape.

Since dropping any tile from  annihilates the benefits mentioned earlier, when compared to yonrenkei or nakabukure¹¹, we tend to keep  and reduce other 4-tile shapes to a mentsu. As shown in Table 20, dropping either of  also gives us far more acceptance than dropping any tile from . Therefore, drop  from (38).

(38) 

When we have 6 blocks, we still tend to keep  shape. When there are only ryanmens in the rest of our hand, drop . In this regard, since there is a weaker  kanchan in (39), we drop

⁹Outside kanchans are kanchans with 4 tiles of ryanmen improvement draws. For pinzu suit, for example, that would be , ,  and .

¹⁰See Section 1.2.1.

¹¹See Section 1.2.2.

Drop			
Acceptance	   12 tiles	  8 tiles	     16 tiles

Table 20: Acceptance of (38).



Drop		
Result shape		
Ryanmen-tenpai draw	8 tiles	8 tiles
Bad-wait draw	6 tiles	8 tiles

Table 22: Acceptance of (43).

The same principle applies to inner shape (ゆうせー, 2022c). Drop from (44) and (45).

(44) East-1, West seat, turn 4, dora

(45) East-1, West seat, turn 4, dora

When we have yakuhai anko such as (46) and draw with shape in our hand, we can drop and keep so that when we pon either or draw we have a good ryanmen wait. Drop from (46).

(46) East-1, West seat, turn 6, dora

Detached Ryankanchan such as 245679

The shape in (47) is dubbed “detached ryankanchan” since it accepts far apart. If we draw we can further improve it into a ryanmenkanchan. On the other hand, dropping either will leave us a mentsu and kanchan; which to drop largely depends on the rest of our hand.

In (47), dropping either of from detached ryankanchan or dropping from will leave us the same number of acceptance at 1-away. However, considering the draws that will give us pinfu and the sanshoku chance, fixing ryanmen here is better, as shown in Table 23 (G・ウザク, 2019, p. 89).

(47)

Choosing between two ryankanchans

(48) has two ryankanchans, one bulging and one vertically developed on one end. Since there is no pair in the rest of our hand, we will focus on how we can make tiles following and pair function. Dropping makes useless since completing shape leaves us no

Drop			
Acceptance	16 tiles	16 tiles	16 tiles
Pinfu draws	8 tiles	4 tiles	4 tiles
Sanshoku-chance draws			None

Table 23: Acceptance of (47).

pair¹². Dropping either of will make full use of all tiles while giving us the same number of acceptance. Taking ryanmen improvement into consideration, drop becomes optimal, as shown in Table 24.

(48)

Drop			
Acceptance	12 tiles	12 tiles	8 tiles
Ryanmen-improvement draws	8 tiles	12 tiles	8 tiles

Table 24: Acceptance and improvement of (48).

When we must choose between two simple ryankanchans, drop the tile that will make our wait better (Feng, 2018, p. 26) Thus, we drop from (49) and make our wait suji. Note that while drop also makes wait suji, is more inside than and thus still has an overall lower win rate.

(49)

1.1.3 134 Shape

It is a common pitfall to only see the ryanmen part of a shape and drop early. The possibilities of can help us tremendously especially when it is still early and our hand lacks shapes (ヨーター, 2024b). In short,

- Makes our hand *kuttsuki*¹³
- Easy to produce a pair

Therefore, when our hand does not have 5 blocks ready, we keep over single yakuhais. When there is no head in our hand, we also prioritize . Otherwise, dropping is fine.

(50) does not have enough blocks, so we keep and expect it to develop. Drop from (50).

¹²Technically, there is, if we treat it as + , but we are then required to drop either of and we are still not tenpai yet despite (48) already at 1-away.

¹³*Kuttsuki* is a Japanese term whose literal meaning is “sticky,” In mahjong context, it refers to a state where tiles next to or 1 tile away from floating tiles all become accepted.

When the 1 is Also a Dora

As shown in Table 26, when we have no akadora in (55), keeping dora  or taking  ryanmen does not make too much difference in terms of hand value. Hence drop  and take ryanmen to increase win rate. When we have one or more akadoras, keeping dora , and possibly calling pon  and pivoting to  tanki or drawing another  and waiting on shanpon greatly increase hand value. When it is still early, with akadoras in our hand, it is better to drop  (ゆうせー, 2022j).

East-1, West seat, dora 

(55) 

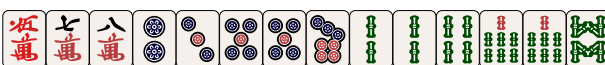
	Drop 	Drop 
 + 	1000 points:  7 tiles; 2000 points:  1 tile	2000 points
 + 	2000 points:  7 tiles; 3900 points:  1 tile	3900 points
 + 	3900 points:  7 tiles; 7700 points:  1 tile	7700 points

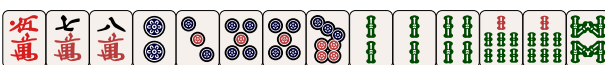
Table 26: Value of (55) depending on drop and draw.

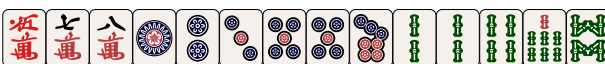
1.1.4 When the 5 of 578 Shape is an Akadora

Hand state	Action
Far away from tenpai	Keep 
1-away	Drop  if tanyao is infeasible
Tenpai	Generally drop 

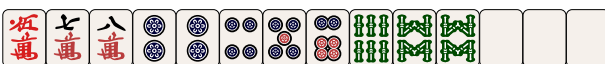
Table 27: Handing .

When we are still far away from tenpai, we can expect to draw tiles around . Since (56) has the risk of ending up no dora bad-wait rii-nomi, we drop  from . (57) also has the same risk, and  has more ryanmen-improvement draws than , so we drop  (ゆうせー, 2022f).

(56) 

(57) 

When we are 1-away, acceptance and final wait become more important than the value the dora brings. (58) already has  anko and we do not have to cling to  for value. Drop .

(58) 

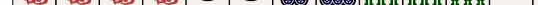
However, as an exception, when keeping  gives us sanshoku or tanyao which enables us to call and speed up our hand, we drop . When reaching tenpai, if dropping  guarantees sanshoku and tanyao, we can also dama our hand instead of dropping  and riichi. In this regard, (59) is

(63) East-1, West seat, turn 3, dora 

There are 6 blocks in (64), which makes us choose between  to drop. If we simply treat  as two kanchans, we are tempted to drop the weakest taatsu, which is  penchan. However, dropping penchan has the following demerits:

-  cannot function as ryanmen (as long as we do not draw )
- We lose  acceptance

On the other hand, if we keep  shape, we will not lose  acceptance because we still have  ryanmen, and drawing  completes a block. The merits of keeping  here, which is to increase the probability of confirming a head, becomes irrelevant due to  pair, so we can drop  without much loss.

(64) 

1.1.6 35689 Shape

In situations where we are choosing between a 3-7 floater and  shape, drop the latter, but not its penchan part. **Generally, drop  and keep penchan at 1-away (ゆうせー, 2022i).** With , riichi chance on  draw is more valuable than forming a taatsu around . Besides, this floater  is inferior to an isolated floater 3 because:

- Drawing  produces a ryanmen with overlapping acceptance
- Drawing  does not give us a ryanmen; it produces a mentsu  and a kanchan 

While it is true that  has  draws that give us ryanmen, from dropping  to drawing any of  upgrade will inevitably take a couple of turns.

That said, there are two cases where we can drop penchan  and go back to 2-away after dropping :

1. Penchan is the bottleneck of tanyao
2. We have other strong shapes such as yonrenkei and nakabukure

When we are comparing between floaters in (65) at 2-away¹⁴,  from  is inferior to isolated . Therefore, drop .

¹⁴While generally we take penchan over floaters at 1-away, when we are 2-away instead, we have some leeway to seek taatsu around floaters and hope for better shapes. Indeed ゆうせー (2020b) states that we can drop penchan if we have two 3-7 floaters at 2-away. However, as stated previously, the ♁ floater is actually weaker than an isolated 3-7 floater, so, in the case of ♁♁♂♂♂♂♂, we do not really meet the standard of dropping penchan.

六萬 八萬

[illegible]

一萬 一萬 三萬 五萬 七萬 四萬 五萬 六萬 七萬 八萬 九萬 十萬 十一萬 十二萬

1.1.8 Shapes with Overlapping Acceptance

shar

Neighboring Pairs



三萬 三萬 六萬 七萬 八萬

三萬 三萬 六萬 七萬 八萬

(73)

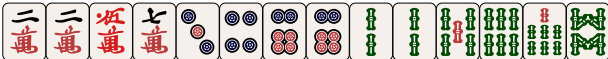
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Skipping Pairs

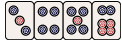
be t

25

 yonrenkei becomes very little. Considering also the possibility of 567 sanshoku, we drop here (G・ウザク, 2016, p. 16).

(77) East-1, East seat, turn 6, dora 


Reaching Tenpai with 3456 Shape

Generally, drop the outermost tile (ヨーテル, 2023b). If we have  and we must dropping one to reach tenpai, dropping  works in general cases. Merits of cutting the outermost tile are:

- Outer tiles generally have less deal-in rate
- Not giving other players easy time calling
- Strength of discards: a  signals that  are relatively safe, while a  gives way less information about our hand.

Exceptions are:

- Doras and sanshoku are involved
- To keep the possibility of catching kamichi's akadora
- To keep possibilities of improvements
- Inner tiles are safer according to table situation
- Early in a round

Cases involving doras and sanshoku

(78) is an open hand with . Our general policy is to drop . However, since dora is  and, if we keep  shape and drawing one , we can drop  instead. This is called sliding and is something one must keep in mind when they have a shuntsu close to dora. Drop  from (78).

(78) Dora 
 

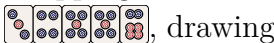
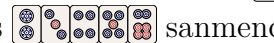
If we drop  from (79), we will not be able to call chi on kamicha's  discard. If we instead keep , we can chi  with  and discard . If  is already discarded, follow the general policy and drop .

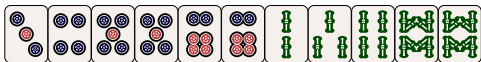
(79)  

(80) has a slight possibility of 123 or 234 sanshoku. In this case, drop .

(80)  

Iipeikou-Like Shapes Has Less Improvements

Dropping either of  from (81) will give us the same waits . If we drop  and keep , drawing  and dropping  gives us  sanmenchan. Dropping  means that we will lose this upgrade. Therefore, drop  here.

(81)  

Note that this is for open hands only since iipeikou shapes gives us no additional value when our hand is open. If this is a closed hand, definitely go for iipeikou since we cannot change our tiles after we riichi.

Half Suji

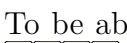
When another player already declared riichi and we must drop either  or  in order to take tenpai, and  is genbutsu, half suji  is safer than musuji  (みーにん, 2017, pp. 140–141). Likewise, when  is genbutsu, half suji  is safer than musuji .

The main reason why half suji is safer than musuji is because, in terms of possibility of the incomplete shape we deal in, musuji  deals into  penchan, which  does not deal into. The number of possible combinations of kanchan waits and ryanmen waits are the same for both , so  has more possible shapes to deal into than , hence higher deal-in rate.

Early in a Round

To make sure that, when we draw  back later in the game we can drop , we can drop  when it is very early in a round.

Cases of 1234 and 2345

To be able to slide when we draw middle tiles, in most cases, we still drop outermost tiles from  or . Other general rules such as keeping sanshoku chance or dora slide still apply. 川村晃裕 (2017, pp. 171–172) also gives similar merits and demerits and suggests to drop innermost tile instead in the following situations:

- Hand is cheap
- We prioritize defense over winning, and 1 (or 2) are generally much safer than 4 (or 5)
- No aka dora acceptance, or aka dora can still be picked up by other blocks
- No tanyao chance
- Shimocha still stays menzen
- We are in huge lead, and we don't mind shimocha opening their hand

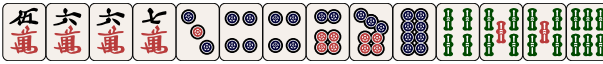
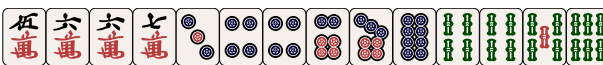
For  however, we should drop  and use  mentsu to catch possible  draw.

1.2.2 Nakabukure

The shape  is called *nakabukure* in Japanese due to the duplicate of the middle tile of the sequence making it look like it is bulging (and hence nicknamed “the bulging shape.”)

Nakabukure works the best with the head of our hand confirmed.  nakabukure accepts , which, except for  draw, make a mentsu and a ryanmen. If we are 1-away and have the head ready, upon drawing any of , we are ready for ryanmen tenpai. Contrarily, if  becomes our final wait, we are forced to wait on  tanki, a very *very* weak wait. Therefore, in cases where we have nakabukure and a ryanmen-toitsu, reducing the latter to a pair often is the best choice.

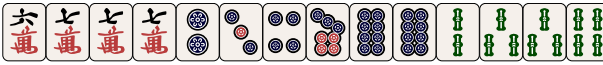
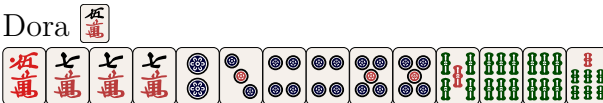
Dropping  from (82) and (83) is the best choice (G・ウザク, 2019, pp. 96–97). Table 30 lists the acceptance of (83) and it shows that confirming our head is better than destroying nakabukure or a ryanmen.

(82)	
(83)	

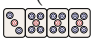
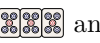
Drop	Acceptance	Good-wait draw
	 38 tiles	 23 tiles
	 37 tiles	 18 tiles
	 33 tiles	 20 tiles

Table 30: Acceptance of (83).

4-tile shapes such as *ryantan*¹⁶ and *kantan*¹⁷ work the best without a head, and thus are incompatible with nakabukure. Since the number of waits of ryanmen and kantan are close to or greater than ryanmens, we destroy nakabukure for better final waits. Therefore, dropping  from (84) and  from (85) are the best choices here (G・ウザク, 2019, pp. 98–99). Especially in the case of (85), even if  is not double dora, knowing that dropping any of  gives us the same number of 34 tiles of acceptance, reducing nakabukure to a mentsu is still slightly better as suggested by Table 31.

(84)	
(85)	

¹⁶Shapes such as  which can be treated as a  ryanmen and  pair, or a  tanki and  anko, hence the name – ryan(men and)tan(ki). As a final wait, it waits on .

¹⁷Shapes such as  which can be treated as a  kanchan and  pair, or a  tanki and  anko, hence the name – kan(chan and)tan(ki). As a final wait, it waits on .

Drop	Good-wait (waiting on ≥ 6 tiles) draw
	21 tiles
	24 tiles
	25 tiles

Table 31: Acceptance of (85).

Nakabukure and a Tile One Tile Away

Shapes such as can be treated as a ryankanchan and pair, or nakabukure and a floating . Therefore, when there are no heads confirmed in our hand, we keep this shape and hope that we get a head with draws.

Drop candidates of (86) are but dropping gives us the widest acceptance as is suggested by Table 32 (G・ウザク, 2019, p. 100).

(86)

Drop	Acceptance	
		20 tiles
		17 tiles
		16 tiles

Table 32: Acceptance of (86). All the candidates share the same acceptance of 11 tiles, which unfortunately give us tanki wait.

In the case of in (87), as is the case of taking headless 1-away, dropping gives us the widest 1-away. In terms of the number of good wait draws, as shown in Table 33, there is only 1 tile of difference between drop and drop. Considering the possibility of improvements with draw, dropping seems better. As long as no yakus or doras are involved, when there is no head confirmed, we prioritize nakabukure and a tile one tile away (G・ウザク, 2019, p. 101).

(87)

Drop	Acceptance	Draws giving us ≥ 6 tiles of tenpai
	 19 tiles	 8 tiles
	 33 tiles	 10 tiles
	 20 tiles	 9 tiles

Table 33: Acceptance of (87).

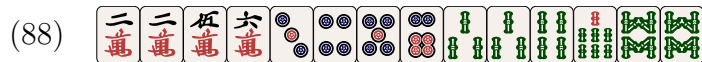
Cases where We Drop Yonrenkei or Nakabukure Before Tenpai

We have learned that both yonrenkei and nakabukure are good at making good shapes. This, however, forces us to treat it as 2 blocks, and to complete 2 mentsus out of these 4-tile shapes, we must draw at least 2 turns. Therefore, when we have enough blocks and they are all sufficient for

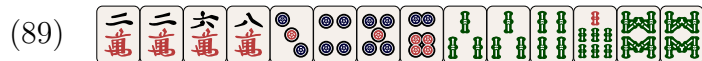
advancing our hand towards tenpai, we can drop yonrenkei or nakabukure without sacrificing too much acceptance. According to クリアレイン (2023d), when there are no other floating tiles to dispose of, the conditions to drop yonrenkei or nakabukure are,

- 2-away
 - Other 4 blocks are all ryanmens and pairs.
- 1-away
 - There is at least 1 ryanmen block.

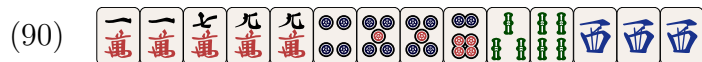
Hand (88) already has 5 blocks (treating yonrenkei as 1 here.) All blocks are good shapes, so we drop either  or  from yonrenkei shape.



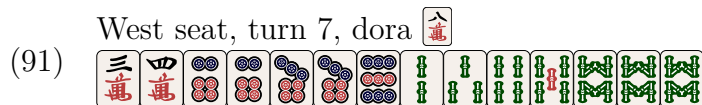
On the other hand, there is a   kanchan in (89). In such cases, we keep yonrenkei or nakabukure in hope for better shapes, while also keeping the bad block so as to not lose its direct acceptance. Therefore, we turn to ryanmen-toitsus and drop . We keep   so that we have a better chance of completing this block with tanyao tiles.



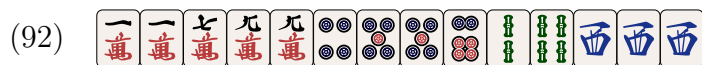
While it is tempting to drop bad blocks and rely on yonrenkei or nakabukure to give us good shapes, when we are at 1-away, we have to compromise and accept the chance of bad waits. Therefore, we do not drop   from (90); we drop  instead.



There is also a   ryanmen in (91), so we drop  from    (多井隆晴・郡道美玲, 2021, pp. 99–100). Dropping  loses the direct tenpai chance on  draw.



However, when the remaining blocks are so bad that there is just no chance for good wait, we can opt to drop one of the bad blocks and fall back on yonrenkei or nakabukure. Therefore, we drop  from (92) and hope for improvement with  draw. We do not touch   since it accepts  6 tiles, which is more than a simple kanchan like  (クリアレイン, 2023d).



1.2.3 Aryanmen

Shapes like are called *aryanmen* in Japanese. The *a*- prefix in *aryanmen* means “next to,” and considering that, as a final wait, the shape has a ryanmen wait but 2 tiles less than an ordinary ryanmen, we can see how this shape is treated as something inferior than a ryanmen.

is also an aryanmen. Since it has no ryanmen improvements, nor is it a strong wait (since we already have two copies of the outermost wait), it is the weakest aryanmen of all shapes in this family. In most cases, it is simply treated as sequence and a floating .

Aryanmen is the Weakest 4-Tile Shape

In terms of good shape improvement, aryanmens have the least number of draws, as shown in Table 34. That is why it is often treated as a weak shape and, in literatures such as Uzaku’s efficiency book, you will find advices such as “when in doubt, drop aryanmen, and more often than not you’ll be right.”

Shape	Acceptance	Good shape draw
Yonrenkei	28 tiles	14 tiles
Nakabukure	16 tiles	14 tiles
Aryanmen	20 tiles	9 tiles

Table 34: Acceptance of 4-tile shapes.

Hand (93) has 3 kinds of 4-tile shapes. We must drop one of them, and we opt to drop the weakest of them all. Drop (G・ウザク, 2019, p. 102).

East round, South seat

(93)

Dropping from (94) and having both sanmenchan and the acceptance of dora confirmed seem good. However, as is shown in Table 35, dropping actually gives us the widest acceptance and the most number of good wait draws.

East-1, East seat, turn 4, dora

(94)

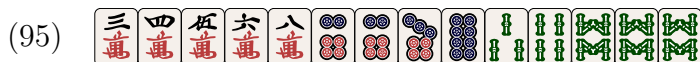
Drop		
Acceptance	51 tiles	23 tiles
Good-wait draws	26 tiles	23 tiles

Table 35: Acceptance of (94).

Let Go of the “Ryanmen” in Aryanmens

Aryanmens does not work well with headless hands. In a 14-tile hand with 2 uncompleted taatsu and 1 aryanmen, if we choose to keep the pair and the ryanmen in the aryanmen shape and drop one tile from other taatsus, at that moment there is one surplus tile from the destroyed taatsu, and thus our acceptance is narrowed. In such cases, it is better to simply reduce the aryanmen to a mentsu. Indeed, headless hands come with the risk of tanki wait. With ankos or other shapes from which we can easily get a head, there is no need to fret about such risk.

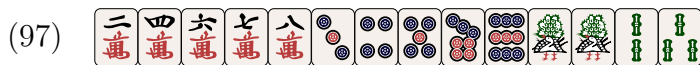
To make all of our tiles useful in (95), we drop  from ayanmen instead of  or . Drawing any of    also gives us ryanmen tenpai (G・ウザク, 2019, p. 105).



Since there is a  pair and a  ryanmenkanchan in (96), dropping  from the aryanmen shape leaves us no surplus tiles. Indeed, dropping  and treating  also as an aryanmen leave us no surplus tile as well, and the difference in acceptance is just 1 tile. However, keeping  aryanmen shape means that we must compromise on  draw which makes us lose tanyao (G・ウザク, 2019, p. 104).



There is 2 kanchan blocks in (97). As shown in Table 36, dropping any tiles from the kanchan blocks leaves us a surplus tile and narrowed acceptance. Taking headless 1-away indeed has the widest acceptance but also the risk of tanki wait, and, in the case of (97), there is no shape that gives us a head or serves as a strong wait when our hand is headless. However, notice that for our  “ryanmen”, we already have two copies of  in our hand, and, if there is another copy of  on the table, this  shape is actually almost as weak as a kanchan waiting on . Considering also the fact that  has no improvement while kanchans at least have ryanmen improvements, dropping  is slightly better here (平澤元氣, 2024).



Drop	Acceptance
	    13 tiles
	      20 tiles

Table 36: Acceptance of (97).

1.2.4 Confirming a Mentsu or Confirming a Head

In cases where our hand is full of ryanmens and shuntsus and the only pair is buried in nakabukure or aryanmen, we must either drop the sequential part or drop the pair part. While it is true that dropping the pair and taking headless 1-away gives us the widest acceptance and thus higher chance of reaching tenpai, we also face the risk of tanki wait. Tanki waits (except for terminal and honor tile tanki waits) have terrible win rate; considering that our goal is to slam that riichi stick

Shape	Confirming head	Confirming mentsu	
	Ryanmen tenpai	Ryanmen tenpai	6-tile tenpai
Ryanmen + ryanmen + shuntsu	16 tiles	12 tiles	0 tile
Ryanmen + sanmenchan	19 tiles	18 tiles	8 tiles
Ryanmen +	15 tiles	18 tiles	8 tiles
Ryanmen +	15 tiles	14 tiles	8 tiles
Ryanmen +	15 tiles	10 tiles	0 tile
Ryanmen (nidouke) + shuntsu	12 tiles	12 tiles	4 tiles

Table 37: The difference between confirming head and confirming mentsu in terms of the number of draws that would give us tenpai with good quality. Bad waits such as shanpon and tanki are omitted here.

with best outcome possible, we would treat this as undesirable outcome¹⁸ and focus on the rest of our acceptance. The theme of this section is thus to maximize the number of good-wait (≥ 6 tiles) tenpai depending on the actual composition of our hand. According to G・ウザク (2019, pp. 46–55) and ゆうせー (2021g),

- Confirming head is better when
 - It is still early and other players are still dealing with useless terminal tiles.
 - We are in a table situation where we are required to force good-wait or yaku.
 - Taatsus are involved in yakus.
- Confirming mentsu is better when
 - It is late and we would like to reach tenpai as soon as possible or to reach keiten¹⁹.
 - Drawing duplicate copies of tiles in taatsus is more likely than completing those taatsus.
 - The shapes of the rest of our hand makes it salvageable even if we complete the taatsus first.
 - Making it a mentsu enables dora slide.
 - We have an anko and thus not required to confirm a head.

In case of the uncompleted taatsus being unrelated ryanmens, headless 1-away has the widest acceptance of 28 tiles, 16 tiles of which complete one of the ryanmen taatsus and force us to drop one tile from the other taatsu and take tanki wait. On the other hand, confirming mentsu leaves us 12 tiles of good-wait tenpai draws, as listed in Table 37. Thus, it generally is better to drop from (98) and confirm our head.

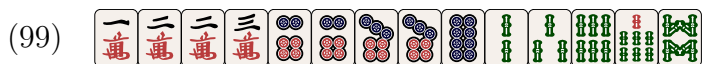
(98)

For shapes like in (99), even if we draw first, we can still drop and make the resulting aryanmen shape as our final wait, not too bad. If this shape sits on the edge like in (100), then we lost the leisure of aryanmen tenpai since does not accept

¹⁸Practically speaking, even if we unfortunately arrive at tanki wait, we can still choose to dama our hand, regardless of it having yaku or not, and wait for our draws that give us better waits.

¹⁹*Keiten*, or *keishiki tenpai*, refers to a state where we are tenpai without a yaku in late turns. At exhaustive draw, players that are not in tenpai must pay those who are in tenpai, so keiten is something we must keep in mind unless someone already called riichi.

“10p.” Therefore, we confirm its mentsu in (99) while we confirm its head in (100).



In the case of in (104), if we drop we will have either aryanmen wait or ryanmen wait on any draws, as shown in Table 38. On the other hand, if we confirm our mentsu with drop, we will have wider acceptance, and more tiles to reach good-wait tenpai. Therefore, drop is the best choice here. Note that the shape becomes a mentsu and a head with draws when our hand is headless.

(104)

Drop		
Acceptance	29 tiles	16 tiles
Good-wait draws	20 tiles	16 tiles

Table 38: Acceptance of (104).

In the case of shape in (105), confirming mentsu here is better since we can either drop or for aryanmen tenpai when ryanmen gets filled; we can dama with draw which gives us iipeiko yaku. Its acceptance is listed in Table 39.

(105)

Drop		
Acceptance	29 tiles	16 tiles
Good-wait draw	20 tiles	16 tiles
Kanchan tenpai	6 tiles	0 tiles
Tanki tenpai	3 tiles	0 tiles

Table 39: Acceptance of (105).

However, if this iipeiko kanchan shape sits on the edge, like shape in (106), confirming mentsu or aryanmen do not differ in terms of good-wait draws, as listed in Table 40. In terms of hand value, however, for the same menzen-exclusive 1 han, iipeiko gives more fu than pinfu. The number of draws that give us yaku after drop is also greater than the acceptance after dropping . Therefore, dropping is slightly better here.

(106)

When the manzu part becomes aryanmen, even though dropping gives us less acceptance, as shown in Table 41, we actually end up 1-away that cannot draw into bad wait. Therefore, the best move here is not to confirm mentsu or to confirm head from aryanmen, but to drop and create two aryanmens.

(107)

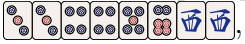
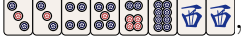
Drop		
Acceptance	 25 tiles	 16 tiles
Good-wait tenpai	 16 tiles	 16 tiles
Kanchan tenpai	 6 tiles	0 tile
Tanki tenpai	 3 tiles	0 tile
Pinfu draw	 12 tiles	 16 tiles
Lipeiko draw	 9 tiles	0 tile

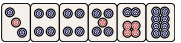
Table 40: Acceptance of (106).

Drop		
Acceptance	 25 tiles	 20 tiles
Good-wait tenpai	 16 tiles	 20 tile
Kanchan tenpai	 6 tiles	0 tile
Tanki tenpai	 3 tiles	0 tile

Table 41: Acceptance of (107).

1.3 6-Tile Shapes

6-tile shapes are a family of shapes that consist of 6 tiles in a suit and, in the context of tile efficiency, compounded with a pair to work with two-pair principle. These particular shapes either are copies of tiles stacked together, like , or have one seemingly useless tile, such as , in a way that often puzzles people and makes them cut tiles that could cost them acceptance and tenpai chance.

Among all the shapes listed in Table 42, the shape  is the most popular one called *ryanmenkanchan* and is the one that is mentioned the most in books and materials on tile efficiency. There are also cases where we have 7-tile shapes such as  or , and the best way of handling this shape depends on many other factors.

Dropping Surplus Taatsus and Forming Ryanmenkanchan

One of the many cases where ryanmenkanchan emerges is when we have surplus taatsus and must drop one. When our taatsus and mentsus are positioned in a particular way, the remaining shape forms a ryanmenkanchan and we get to keep part of the acceptance of the destroyed taatsu, thus minimalizing our loss. Examples include:

- Dropping  from 
- Dropping  from 
- Dropping  from 

In (108), if we drop either of , at that moment we have a surplus tile in our hand, and we end up 4 kinds 16 tiles of acceptance, as shown in 43. The key here is to identify the  ryanmenkanchan, which makes use of all souze tiles and gives us one more kind of

Shape	Acceptance	Improvement
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop
		Draw drop

Table 42: 6-tile shapes, with a head included (chitist, 2021).

tiles of acceptance. Drop (G・ウザク, 2019, p. 118).

(108)

Drop		
Acceptance		

Table 43: Acceptance of (108).

(109) has 6 blocks consisting of mentsus, ryanmens, and pairs. We intend to drop either of the pair. The pinzu shape may look completed, but if we drop there is actually ryanmenkanchan in our hand and it gives us more acceptance than dropping .

(109)

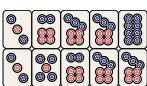
(110) is a similar case where dropping gives us the ryanmenkanchan which makes full use of every tiles (ヨーテル, 2023f).

East-1, turn 4

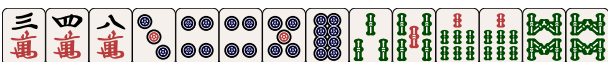
(110)

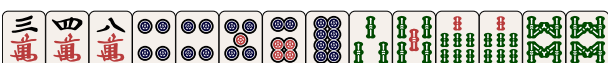
Floating Tiles and Ryanmenkanchan Potential

When comparing floating tiles in our hand, we tend to overlook some floating tiles that position near other shapes in a way that has ryanmenkanchan potential. For example,

-  in want of 
-  in want of 

Both (111) and (112) have floating  and , and  is more isolated than , so we drop  in hope that we can get ryanmenkanchan in pinzu suit (G・ウザク, 2019, p. 122).

(111) 

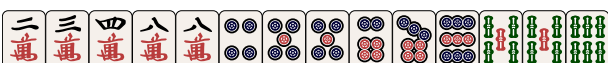
(112) 

Choosing between Pinfu and Tanyao

For the same 1 han, tanyao has a probability of scoring higher than pinfu, and we are ready to make calls and speed up, so we prioritize tanyao over pinfu (G・ウザク, 2019, p. 119). Dropping either of  from (113) leaves us no surplus tile and gives us the same 19 tiles of acceptance. Dropping  gives us tanyao chance, and dropping  confirms pinfu. Considering the merits mentioned earlier, we opt to drop  and go for tanyao.

(113) 

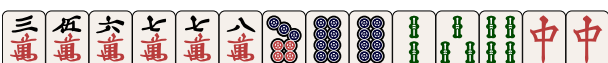
Likewise for (114), we drop  and choose tanyao (G・ウザク, 2016, p. 10).

(114) 

Cases where Pinfu is Impossible

Due to the characteristic of ryanmenkanchan that it always produces two shuntsu when completed, it works very well with pinfu. For cases where pinfu is not within sight such as having yakuhai pair in our hand, the importance of maintaining ryanmenkanchan decreases.

Hand (115) has  pair and thus cannot have pinfu. Dropping  and  do not have any difference in terms of acceptance, but if we drop  here, we end up with just rii-nomi. Therefore, we drop  here. We trade  acceptance with  acceptance (ヨーテル, 2022c).

(115) 

It does not matter too much which to drop for rii-nomi hands such as (116). Adjust according to actual table situation (ヨーテル, 2022c).

(116)

Choosing between Guaranteed Pinfu and Iipeiko Chance

For shapes like or , we can either make it ryanmenkanchan, or make it other kinds of 6-tile shapes. Either way, we still have 11 tiles of acceptance with the remaining shape. Therefore, the difference lies in the value the remaining shape could bring us.

Drawing with or in (117) or (118) gives us iipeiko, so, in both cases, dropping gives us the most possible value. It is true that, since the remaining shapes after drop all have shanpon acceptance, we still could end up rii-nomi. However, The increase in value with one more han is greater than the decrease in value with one less han, so, in terms of expected value, drop is still optimal (G・ウザク, 2019, pp. 120–121; ヨーテル, 2022c). The cases where we would choose ryanmenkanchan and guaranteed pinfu instead are,

- *Agaritoppu*, a kind of table situation where winning our hand confirms our first place.
- In lead, and we are going to crush others' renchan²² chance
- Already valuable, such as dora 3.

(117)

(118)

Drop		
Acceptance	                          	

Table 45: Acceptance of (120).

Ryanmen-Toitsu and 6-Tile Shape

When we are choosing between ryanmen-toitsu and 6-tile shape (with also a pair in our hand,) reducing ryanmen-toitsu to a ryanmen is, in most cases, the best choice in terms of acceptance and, when the number of accepted tiles are the same, the number of good-wait draws. Hand has which may seem weak, but it indeed is a 6-tile shape according to Table 42. Dropping and have the same acceptance of 16 tiles, but, as shown in Table 46, keeping has the maximal 8 tiles of good-wait draws. By dropping , we are actively denying the possibility of draws that destroy our ryanmen.

(121)

Drop	Acceptance	Good-wait draws
	16 tiles	4 tiles
	14 tiles	4 tiles
	16 tiles	8 tiles

Table 46: Acceptance of (121).

(122) too follows the same principle and drop gives us the maximal acceptance. Even though sits on the edge and loses 4 tiles than its middle counterpart, in terms of total acceptance, drop still has more acceptance than, say, drop (G・ウザク, 2019, p. 134).

(122)

Drop	Acceptance
	20 tiles
	17 tiles
	14 tiles

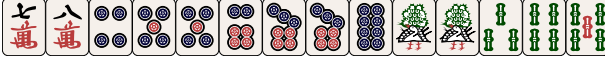
Table 47: Acceptance of (122).

From 7 Tiles to 6 Tiles

In the following cases, we will deal with 7-tile shapes that give us possibility of different 6-tile shapes. The point is:

- Always pick the 6-tile shape that gives us the maximal acceptance.
- If the result 6-tile shapes have the same number of acceptance, pick the one with most yaku potential, be it tanyao, pinfu, or iipeiko.

We must drop one tile from pinzu part of (123). We can either drop any of  and have what is called “sanmen-toitsu,”²³ or drop any of  and have what is called “*niinii* shape.”²⁴ Either way, we have  acceptance, so the difference is in the acceptance of sanmen-toitsu and niinii shape. As is shown in both Table 42 and Table 48, niinii shapes have 4 less tiles than sanmen-toitsu, so the correct drop is . If  is still live, drop  to accept it (G・ウザク, 2019, p. 132).

(123) 

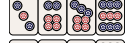
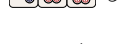
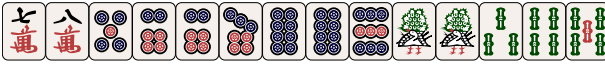
Drop	Acceptance of pinzu shape
	 13 tiles
	 13 tiles
	 9 tiles
	 9 tiles

Table 48: Acceptance of (123).

When the shape sits on the edge like in (124), the sanmen-toitsu shape loses 4 tiles of acceptance and there is no difference, in terms of acceptance, between  drop and  drop. However, by keeping  niinii shape, we have iipeiko on  draw and thus the highest expected hand value. Therefore, drop  (G・ウザク, 2019, pp. 132–133).

(124) 

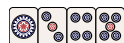
Shapes like  accepts 1 less tile than sanmen-toitsu but still 3 more tiles than niinii shape. Therefore, we drop  from (125) for maximal acceptance. If  becomes  and loses 4 tiles of acceptance as in (126), taking  niinii shapes instead gives us the maximal acceptance and also iipeiko chance (G・ウザク, 2019, p. 133).

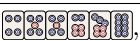
(125) 

(126) 

1.4 Other Shapes

1.4.1 1345 Shape

Keep  in  over isolated  (noprops, 2022). Drawing  gives us  ryanmen, and drawing  gives us a  kanchan which can be a decent wait. Compared to an isolated  which produces a penchan if we draw , kanchan if we draw , and a ryanmen if we draw ,  has slight advantage. In this regard, we drop  from (127).

²³A name I made up since, for , it is a sanmenchan  and a toitsu .

²⁴From Japanese where, for , *ni* “two” refers to  which we have two copies and *i* “one” refers to  which we have one copy.

East-1, West seat, turn 3, 

1.4.2 13457 Shape

Keep isolated  **over**  **in**     (noprops, 2022). Drawing  with     makes  much weaker since, if we drop , we will not lose sanmenchan chance with  draw, and we can still incorporate  since    is a shape that has ryanmen-improvement draws. Therefore, drop from (128).

(128) East-1, East seat, turn 4, dora 

1.4.3 Entotsu

The shape  is called *entotsu* in Japanese whose literal meaning is “chimney.” Why this particular shape is associated with chimneys remains disputed. Since  can be treated as either  +  and  + , with  pair in our hand, we end up  wait.

(129) is an open hand and we draw . Since dora is , we would like to work our shapes closer to dora. On top of that, since we have a kanchan  in our hand, it is more likely to draw another copy of  (6 tiles) than tsumo our winning tile ; in the former case, we can upgrade our hand to entotsu shape and catch dora . Therefore, drop .²⁵

(129) South-2, South seat, turn 4, dora

(130) has 6 blocks (seeing pinzu part as $\begin{bmatrix} \text{red} & \text{red} \\ \text{green} & \text{green} \end{bmatrix} + \begin{bmatrix} \text{red} & \text{red} & \text{red} \\ \text{green} & \text{green} & \text{green} \end{bmatrix}$) and dropping any taatsu will not cause too much difference – any of $\begin{bmatrix} \text{red} & \text{red} \\ \text{green} & \text{green} \end{bmatrix}$ drop gives 16 tiles of acceptance and any of $\begin{bmatrix} \text{red} & \text{red} & \text{red} \\ \text{green} & \text{green} & \text{green} \end{bmatrix}$ gives 15 tiles. According to Section 1.5.2, the best course of action is to drop a pair. Although there is no immediate difference between $\begin{bmatrix} \text{red} \\ \text{green} \end{bmatrix}$ drop and $\begin{bmatrix} \text{red} & \text{red} \\ \text{green} & \text{green} \end{bmatrix}$ drop, but keep in mind that $\begin{bmatrix} \text{red} & \text{red} & \text{red} & \text{red} \\ \text{green} & \text{green} & \text{green} & \text{green} \end{bmatrix}$ can become shapes listed in Table tab:6plus2 with one additional pinzu tile, so $\begin{bmatrix} \text{red} \\ \text{green} \end{bmatrix}$ drop is the best here (G・ウザク, 2019, p. 130).

(130)

(131) is similar to (130) but with its shapes more to the middle and one additional anko, tanyao and sananko weigh than pure efficiency and the improvements with future draws. Since  is now our bottleneck when it comes to scoring tanyao, we can drop it here (G・ウザク, 2019, p. 131).

(131)

²⁵This is from one of my games and in the end I did pick entotsu shape and draw .

1.5 6 Block Theory

We have seen the term “block” here and there previously without defining its exact meaning. In this section, we will cover the term “block” thoroughly and explore the possibility of keeping multiple blocks in our hand, even cases where we have more blocks than needed.

A block is a group of tiles that we expect to become mentsu when we complete our hand. To complete a hand in mahjong using 14 tiles, we need 4 mentsus and 1 head²⁶. In that regard, we may think that we only need to keep 5 blocks in our hand. However, when we are still far away from completion and each block does not take up too much space, we can fit 6 blocks in our hand. The method of developing our hand while keeping all 6 blocks is called “6 block theory” (G・ウザク, 2019, p. 16).

Merits of keeping 6 blocks include (G・ウザク, 2019, p. 17; ゆうせー, 2022g; ゆうせー, 2022m):

- Keeping maximal choices between yakus and waits
- Able to adapt to new table situation such as pon or kan from other players
- Dropping tiles and fixing ryanmens make their outside wait come out more easily

Demerits of keeping 6 blocks include (G・ウザク, 2019, p. 18; ゆうせー, 2022g):

- Keeping 6 blocks leaves us no room for follow-up tiles for each block, thus harder to reach perfect 1-away
- The moment we drop 1 block to take 1-away, the destroyed block leaves us 1 surplus tile, thus making our acceptance narrower and lowers our defense

1.5.1 Criteria of Keeping 6 Blocks

Our policy regarding block number in our hand is (G・ウザク, 2019, p. 22–23; ヨーテル, 2023c; ゆうせー, 2022g; ゆうせー, 2022m):

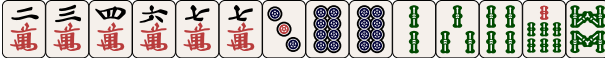
- Keep 6 blocks when
 - Going straightforward leaves us high probability of cheap bad wait
 - Blocks involve hand value and yaku
 - There is no huge difference between blocks
- Keep 5 blocks when
 - There is huge difference between blocks
 - We prioritize the speed of reaching tenpai
 - Our hand is open and we want pairs for pon calls
 - We prioritize defense

Cases where We Keep 6 Blocks

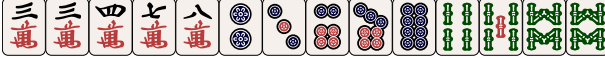
Keep 6 blocks if going straightforward with blocks we currently have leaves us bad waits or cheap hand (G・ウザク, 2019, p. 23). For (132), if there is no difference in player points, drop  and take perfect 1-away. However, if we are far behind other players, keeping 

²⁶Except for kokushi musou and chiitoitsu.

gives us a slight chance of sanshoku and dropping  confirms pinfu. This is the case where even a single tile can be treated as a whole block.

(132) 

When there is no discernible difference between taatsus, we can delay our decision by keeping 6 blocks (ヨーテル, 2023c). (133) has 6 taatsus, and dropping any of  leaves us 5 blocks. We can also reduce  to a ryanmen and progress with 6 blocks.

(133) East-1, South seat, turn 3, dora 


We can drop  from (134) and keep 6 blocks, or drop  and reduce it to 5 blocks²⁷. If we drop , drawing  will not destroy sanshoku, and we keep the possibility of drawing another copy of . If any other ryanmen gets filled up before we complete , we can drop  and aim for pinfu. This is the case where keeping 6 blocks gives us more choices between yakus (ゆうせー, 2022g).

(134) East-1, South seat, turn 3, dora 

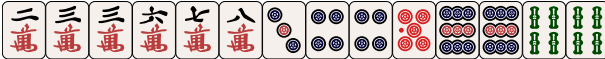

There are 3 possible options in (135),

- Drop  and wait for improvement of 
- Drop  and follow pure tile efficiency
- Drop  and force tanyao

Considering the current turn and the value of our hand,

- It is already in the middle of a round → we cannot afford going back in shanten with  drop.
- There are 3 pairs and we could possibly end up rii-nomi without tanyao → dropping  is not favorable.
- It is more easily to make another group out of  than out of .

Therefore, dropping  is the best choice here (福地誠, 2017, pp. 57–58). If we draw around , we can drop  and go for tanyao. If we draw  first, we can still drop  and riichi.

(135) Turn 8, dora 


²⁷If we drop  we lose pinfu and sanshoku becomes uncertain. Do not drop .

Cases where We Keep 5 Blocks

(136) has a penchan that is clearly weaker than kanchan, so we drop 石橋伸洋 (2015, p. 21).

(136)

(137) has  shape which can be treated as 2 blocks  + . If we reduce it to  1 block, we have the widest acceptance of 7 kinds 24 tiles. However, if we don't draw any of  9 tiles (1 copy of  is the dora indicator) first, we end up a weak penchan wait. We therefore drop . If we later draw another copy of dora , we can drop  (G・ウザク, 2016, p. 16).

(137) East-1, East seat, turn 7, dora 



Being able to pon is crucial for open hands (ヨーテル, 2023c). We just called pon  in (138). 6 block theory would require us to drop  here and abandon the ability to call  pon, which defies the purpose of going open in the first place. Open hands also come with the ability to change tiles and waits, so having tiles following taatsus becomes more important and contributes to our flexibility. If (138) develops into (139), calling  chi gives us two ryanmens, drawing  also improves our hand into ryanmens. Therefore, drop either of  depending on table situation.

(138) 

[illegible]

At the moment we draw  in (140), we have 6 blocks. The only purpose of   is to accept ; if the other taatsus get completed first, we end up  penchan wait rii-nomi. It is therefore close to useless, and we would expect to draw another copy of   to give our hand more value; even for defense purposes,   is generally more safe than  . Thus we drop   here (お知らせ, 2020, pp. 154–155).

(140) East 1, East seat, turn 1, dora 

(141) already has 5 blocks necessary for manzu honitsu. If we keep , at worst our hand becomes haku nomi 1500 points. Therefore we drop . Keep in mind that dora is  and keeping  makes us able to incorporate dora (お知らせ, 2020, pp. 156–157).

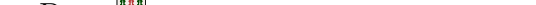
(141) East-1, East seat, turn 4, dora

The hand consists of:
- Four red tiles: 三萬 (Sanman), 三萬 (Sanman), 伍萬 (Gomwan), 七萬 (Shichiman).
- Two blue tiles: 二筒 (Nigō), 四筒 (Yōgō).
- Three green tiles: 一索 (Issō), 二索 (Nisō), 三索 (Sāso).
- Two yellow tiles: 東 (Tō), 東 (Tō).
- Two purple tiles: 北 (Kai), 北 (Kai).
- One green tile: 發 (Hatsu).

Comparing Different Shapes

(142) 

(143) Dora 



- By keeping two pairs, we keep the possibility of their ryanmen improvements and therefore the possibility of perfect 1-away
- If our pairs are outside, they come out more likely, especially for honor tile pairs
- Two pairs means that our waits becomes *atosuji*²⁸ twice likely
- Pairs do not use neighboring tiles, thus making them no-chance²⁹ more likely

Even for pairs like which lack any ryanmen improvement, due to them being the outmost number tiles and twice likely becoming atosuji than a kanchan, they are decent as final waits. In this regard, the only cases where we would keep kanchan is when we have yakus such as tanyao or sanshoku involved, or when table situation makes pairs less favorable than kanchan.

²⁹If there are 4 copies of , it makes  wait or  wait *no-chance* since the ryanmens waiting on  or  requires 1 copy of . It makes  kanchan also impossible. It, however, does not negate the possibility of   or   shanpon wait.

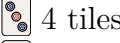
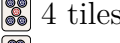
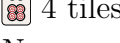
Shape		
Acceptance	 4 tiles	 4 tiles
Ryanmen improvement	 4 tiles	 8 tiles
Ryankanchan improvement	 4 tiles	None
Kanchan-toitsu improvement	None	 8 tiles

Table 49: Comparing  and .

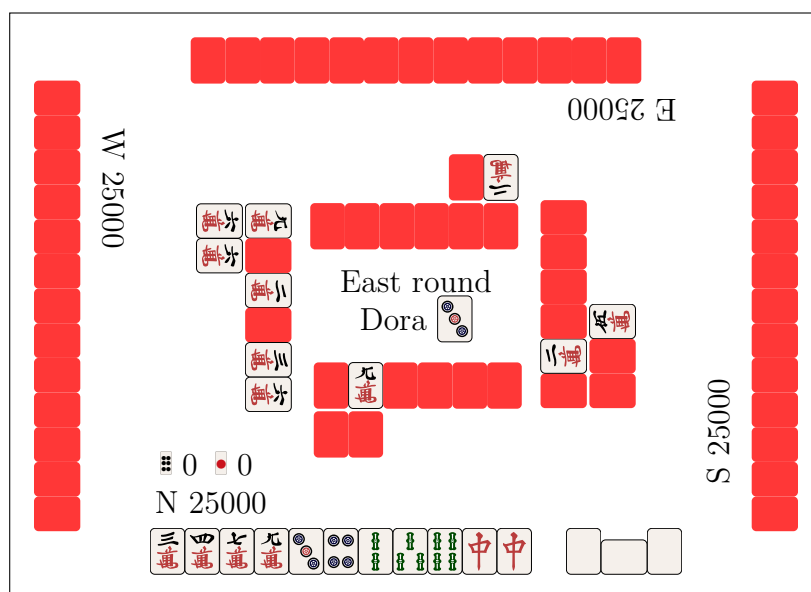
(お知らせ, 2020, pp. 152–153). At the moment of drawing  in (144), we have 6 blocks. We will not touch yakuhai pairs that give us value and speed; in that regard, we have no incentive to keep . Instead of keeping  and maintaining 6 blocks until we complete another block, we must drop it immediately to make room for  safe tile. Drop .

(144) South-2, East seat, turn 4, dora 


When we are comparing ryanmens and kanchan, when they have the same number of acceptance left, **keep kanchan** (福地誠, 2017, pp.148–149, p. 218). In Situation 1, a lot of manzu tiles has been cut, and both of  ryanmen and  kanchan in our hand accepts 4 tiles. Here, kanchan is better than ryanmen because,

- We can draw  and upgrade it into a ryanmen.
- Even if all copies of  are cut, kan  becomes no-chance and more likely to come out.
- Occasional increase in value due to the fu given by kanchan.

Even though 234 sanshoku seems plausible, with 3 copies of  already cut, and  ryanmen still uncertain which side of it is getting filled up, we opt to drop  here.



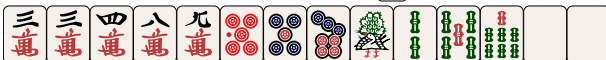
Situation 1

Comparing Similar Shapes

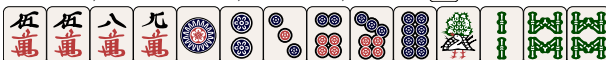
Even for shapes in the same class, since their acceptances are same (without table condition,) their differences lie in their improvement and shapes around them. In particular, we must also take into consideration how bad it backfires when we drop these taatsus. For each class of shapes, we have following merits (お知らせ, 2020, pp. 152–153; ゆうせー, 2022l):

1. (Kanchan, penchan) remaining number of acceptance
2. (Ryanmen, penchan) follow-up of acceptance
3. (Kanchan) follow-up of ryanmen-improvement, ryankanchan-improvement draws
4. (Toitsu) whether or not we can slide towards dora
5. (Toitsu) whether or not, when reduced to a floater, it has improvement

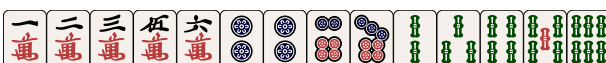
There are two penchans  and  in (145). To prepare for the moment we pon , we must keep another pair , so we cannot drop . In this hand, after dropping , its acceptance  can still be incorporated by its neighboring kanchan, which forms  ryankanchan. On the other hand, drawing  after dropping  is useless; it is just a floater with furiten risk. In other words, dropping  loses the least. Drop  (ゆうせー, 2022l).

East-1, North seat, turn 3, 
 (145) 

(146) is also choosing between  and . Since we have  pair, the acceptance  of penchan  gives us  kanchan-toitsu, and its kanchan-improvement draw  gives  ryanmen-toitsu. When there is no other information available to us, judging on how we can still incorporate accepted tiles and improvement draws,  becomes the weakest penchan. However, when  is once cut, or when we immediately draw  after dropping , we still keep manzu and drop  (ゆうせー, 2022l).

East-1, East seat, turn 2, dora 
 (146) 

(147) is choosing between  and . With  shape, when we drop  and draw , we can still slide towards  and drop . Drop  (ネマタ, 2014j).

(147) 

(148) is choosing between  and . With  shape, when we draw another dora , we can drop  and maintaining a head in our hand. Therefore, drop  (G・ウザク, 2016, p. 20).

East-1, West seat, turn 7, dora 
 (148) 

1.5.3 Dealing with Multiple Pairs

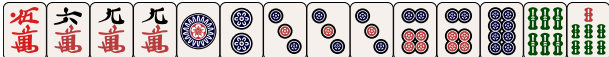
We have discussed situations where we would destroy pairs or keep them in Section 1.1.1. However, the arguments in Section 1.1.1 are made in the context of 5 blocks. In the following section, we will look at pairs in the context of 6 block theory.

When there are three pairs, it is generally a good idea to destroy one (ネマタ, 2014d), **since keeping pair has less acceptance than keeping ryanmen or kanchan** (p. 34 G・ウザク, 2019). We value the strength of a pair according to the following merits (ネマタ, 2014d; G・ウザク, 2019, p. 169; G・ウザク, 2016, p. 8),

- The number of improvement draws
- Involvement in yakus
- Strength as final waits

There are three pairs in (149) and we must drop one. Considering that we can get  shape with  draw,  becomes the most useless pair. Drop .

(149)

(153) East round, non-dealer, dora 


2 Building Yaku

Most of Section 1 is about choosing the choice that gives us the most tiles of acceptance with good quality. In cases where choices lead to the same number of acceptance, we then start discussing the expected value of our choices. In this section, we are going to focus more on the value of our hand, and, to some extent, to defy tile efficiency and choose actions that will not give us the most acceptance.

In a ruleset with yakus such as dora, akadora, ippatsu, and uradora, players can easily reach mangan without putting too much thought in building their hand; all they have to do is to be a little bit luckier with how the wall is built. In a ruleset where these opportunistic yakus do not exist, or when we are in dire need of points, making rii-nomi hands seems equal to digging our own grave, and we certainly do not want to sacrifice value potential just to reach rii-nomi faster.

Mahjong is a game of building mangan. For 30 fu hands, we start at 1000; for each additional 1 han we have, our hand value doubles, until it reaches 7700 for 4 han (or 8000 if it rounds up in some rulesets.) Likewise for other hands with different fu numbers, our hand value doubles as han number goes up until it reaches mangan. Therefore, when we have 1 or 2 han confirmed in our hand, spending effort on building hand value becomes worthwhile.

When we already have 3 han or more, yakus are still important – it gives us the opportunity to dama or go open. Since the value of our hand is close to mangan cap, the benefit of riichi (potential ippatsu and uradora) becomes marginal, which then does not justify the decrease in win rate. Only riichi as last resort – no yaku, and must push.

For a concrete example of how much we can sacrifice our efficiency for value, see (154). If we drop , as is prescribed by tile efficiency, we end up 3900 points. Unless we are in huge lead in south game, we can opt to drop , and keep ryanmen ryanmen for 7700 (福地誠, 2017, pp. 98–99).

(154) Turn 8, dora 


2.1 Yakuhai

Yakuhais refer to a family of honor tiles that gives us 1 han when we have 3 copies of them. Yakuhais include,

- Sangenpai: any of haku , hatsu , or chun .
- Bakaze: the same wind as the round. For east games, it would be .
- Jikaze: the same wind as your own seat. For a player in South seat, it would be .

Honor tiles that are not included in the previous categories are called *otakaze*.

When we have a pair of yakuhais in our hand, we can call pon when one copy of it is dropped and we have a yaku confirmed instantly. When to call pon or not will be left for the section of call judgement. In this section, we are going to deal with lone yakuhai tiles and compare it with other floaters.

To put it simply, **it depends**. While we always teach beginners to drop honor tiles first, there are certainly situations where we would keep honor tiles and drop 19 or even 28 floaters first. Efficiency wise, honor tiles can only accept same kinds of tiles, so their acceptance always is less than a floating 1 or 9. However, the value and potential increase in speed it brings us when it pairs up makes it more versatile than a floating 1 or 9, whose job is to either provide a head or to create a penchan or kanchan.

The assessment of pure tile efficiency and need of speed depends on table situation, hand value, hand shape, and of course the person making the assessment. Different people and AIs have their own rules and preferences. The list below shows different kinds of preferences from 6 sources.

- 千羽黒乃 (2020)
 - Hand is in good shape (one pair and two good shape blocks), 19 floater > otakaze > yakuhai
 - Otherwise, yakuhai > otakaze > 19 floaters
- Maru-Jan チャンネル (2014)
 - Hand close to winning, or hand we want to win: 19 floaters > otakaze > yakuhai
 - Hand that does not seem close to winning: otakaze > yakuhai > 28 floaters > 19 floaters
- NAGA's preference according to 美咲 (2020)
 - Pinfu hand: otakaze > yakuhai
 - Otherwise: yakuhai > 19 floaters > otakaze
- 渋谷難波 (2023b)
 - The 1 in 1345 > yakuhai > otakaze
 - Otakaze will be your first drop most of the time
- ヨーテル (2021b)
 - One dora: 19 floaters > yakuhai > otakaze
 - No dora: yakuhai > 19 floaters > otakaze
- 稚児 (2023)
 - For headless hands, drop yakuhais first.

To sum these up, it would be three kinds of orders, depending on our hand:

1. It is slow to remain closed, or it would very likely end up valueless rii-nomi: yakuhai > otakaze > 19 floaters
2. Pinfu or tanyao hands, or headless hands: 19 floaters > otakaze > yakuhai
3. Otherwise, hand is not too good nor too bad: 19 floaters > yakuhai > otakaze

Comparing Yakuhais and Floaters

A more detailed comparison between yakuhais and 19 and 28 floaters is as follows (ゆうせー, 2024a; ゆうせー, 2023a; ZERO, 2020, pp. 14–15),

- Keep yakuhais over 19 floaters when
 - Hand is too far away from tenpai when it stays closed (the number of ryanmens and mentsus are no more than 2.)
 - The floater is suji to another floater, such as .
 - We don't have enough blocks and we have a pair.
 - We have 5 blocks but one of those is bad shape.
 - Pinfu does not seem doable.
 - Floaters are not related to dora.
- Keep yakuhais over 28 floaters when
 - There are more than 2 bad shapes and no completed mentsu.
 - Terminal tiles are dora.

For hands such as (155) which only has two ryanmens, it would take long to complete the kanchans or to improve them, not to mention  still could give us penchans; by the time they are filled up, our opponents already tsumo'd their riichi. Keeping yakuhais here not only potentially speeds up when they pair up, but they also are generally safer than number tiles. Therefore, we opt to drop either of  here (ゆうせー, 2024a).

(155) East-2, South seat, turn 1, dora 

																						
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Hand (156) has only 4 blocks. Efficiency wise, we could use  acceptance of  floater. However, drawing a taatsu around  means that  becomes surplus, and we end up dora 1 kanchan/penchan riichi of 2600 if we cannot improve the taatsu in time. Value wise,  is the most useless tile; we would like to see draws near , or drawing another copy of  to give us good wait 2000 (ゆうせー, 2024a).

(156) East-1, East seat, turn 3, dora 



Since there is only 1 ryanmen and 1 mentsu in (157), it meets the criteria of keeping yakuhai over terminal floaters and the choices are  and . When we drop , its acceptance  cannot be picked up by its neighboring taatsu  . If we drop ,  forms a ryankanchan with   and  is the ryanmen improvement of that kanchan. Therefore, we drop  which backfires the least (ゆうせー, 2024a).

(157) East-1, South seat, turn 4, dora 



三萬 四萬 五萬 六萬 七萬 八萬 九萬 東 發 中



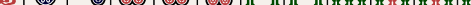
eat

四萬 四萬 七萬 四四 五五 六六 紅中 紅中 紅中 紅中 紅中 紅中 北 北 北 發 中

東發中，



二萬 六萬 八萬

[illegible]

Yakuhei Attack

Yakuhai attack refers to the strategy where we drop yakuhais first before they get to pair up in others hands, or when others hands are still in a state unready to call (ZERO, 2020, pp. 16–17).

Dropping yakuhai first means that our hand is blessed with good shapes, or yakuhais contribute little to our hand. In other words,

- Our hand is full of shuntsus or taatsus and there are no pairs; calling pon on yakuhais does not speed up our hand.
- Our hand consists of only tanyao tiles if we drop yakuhai tiles.
- Our hand is “in good shape,” and to be in good shape, it has to
 - Have more than 3 ryanmens or mentsus combined (ZERO, 2020, p. 17)
 - Have 1 pair and two good shape blocks (including kanchan-toitsus) (千羽黒乃, 2020)

If our hand meets any of the aforementioned conditions, we can drop yakuhais before otakaze. Note that otakaze can serve as our head for pinfu. For example, for hands like (164) where tanyao is very likely, or for hands like (165) which is in good shape according to 千羽黒乃 (2020)³⁰, we can drop yakuhais first.

(164) East round, West seat, turn 1



(165) East-1, North seat, turn 1, dora 一萬



We already have two yakuhai pairs in (166), more than enough to end the game and take first place. It means that 中 does not contribute to our victory and, if we hold it for too long and drop it too late, we may end up letting others call it. Therefore we drop 中 here first despite an otakaze 西 in our hand (川村晃裕, 2017, pp. 181–182).

(166) All-last, non-dealer, haipai, dora 北, 500 points to overtake first place



Hand (167) is the moment we pon dora 一萬. The lone yakuhai 中 again contributes very little to our hand, since its value is already mangan. At this rate, the demerit of letting others call and possibly allowing them to win before we could is greater than the merit of value or defense. Drop 中, even though 一萬 is close to useless also (川村晃裕, 2017, pp. 183–184).

(167) East-1, non-dealer, turn 1, dora 一萬



³⁰ 一萬 would be the first to go according to ゆうせー (2024a): not enough blocks, having one pair. Some see hands like this doable, some see this far away from tenpai. This is why different folks have different preferences when it comes to yakuhai and terminal tiles.

Lone Dora Yakuhai

If there are other yakus for our hand, drop lone dora yakuhai before anything else. **Otherwise, do not drop it easily** (ヨーテル, 2022a). As mentioned earlier, if our hand already has strength points such as tanyao or pinfu, we do not particularly need dora yakuhai to pair up, and dropping them early on reduces the probability of them getting called pon. Therefore, for hands like (168) with tanyao and dora and pinfu potential, we drop  first.

(168) Dora 



For (169), however, it has no pinfu, no tanyao, and no dora whatsoever, so we keep  and other honor tiles. Drop any of         .

[illegible]

For (170) which is *mentanpin*³¹ 1-away, we would still drop *dora*  first, even if we draw this scary thing this late.

Turn 7, dora

(170) 

If our hand already has mangan value, it is better to drop lone dora yakuhai. (171) already has honitsu aka and chun, which is 8000, and getting  paired up only gives us 4000 extra points. If we cannot get it paired up in time and someone calls riichi, it would be very hard to drop it against riichi. Keep it if you are falling behind and are in dire need of points.

(171) East-1, South seat, turn 7, dora 

2.2 Tanyao

Dropping Terminal Block from 6 Blocks

In most cases, keeping 5 tanyao blocks and drop the non-tanyao one is correct(ネマタ, 2014e; ZERO, 2020, p. 22), even if it is not the choice that gives us the most acceptance. If we drop either  or  from (172) we will have maximal acceptance of 16 tiles. However, we will only get pinfu with  draw; any other tiles will only give us rii-nomi. If we drop , we have tanyao confirmed, and we can open our hand. Drop .

(172)

³¹Short for menzen (riichi), tanyao, and pinfu.

East-1, South seat, turn 3, dora 

East-1, South seat, turn 3, dora 

Turn 4, dora

East round, North seat, turn 4, dora 

East-1, South seat, turn 3, dora 

56

; drop  if we are in the middle of a round for safety reasons, and drop  for possible iipeiko improvement with  draw.

(178) East-1, South seat, turn 3, dora

Likewise, for (179), we go full efficiency and drop the entire ryankanchan. Although this hand looks like (173), since (173) is too far away from menzen tenpai, we had to force tanyao. For cases like (179), we are closer to menzen tenpai with 3 ryanmen blocks and not in need to force tanyao.

(179) East-1, South seat, turn 3, dora

Likewise, forcing tanyao in (180) by dropping  leaves us pairs and kanchan and makes us lose the acceptance of dora . Efficiency wise, keeping two pairs is better than a kanchan and a pair in terms of possible ryanmen improvement draws, so we drop  here (福地誠, 2017, pp. 32–33).

(180) Turn 5, dora 



Forcing Tanyao when Agaritoppu

If we are agaritoppu, we must win our hand as soon as possible, and tanyao becomes our most reliable tool. For abysmal hands like (181), we are forced to call, and  penchan becomes completely useless that we can drop  first. After that, if we draw any of , , drop . We can also call any of  and drop  (in the case of  chii, we cannot drop ; drop  instead.) (川村晃裕, 2017, pp. 11–12).

(181) All-last, West seat, turn 1, dora , 500 points to overtake first place

Hand (182) is hopeless, and we still want to get that first place. As always, tanyao is our most reliable tool; yakuhai is possible too. In this regard, since we already have 4 tanyao blocks, the 此 pair is the most useless. Drop 此. Same for (183) which has two ryanmens (川村晃裕, 2017, pp. 27–30).

(182) All-last, South seat, turn 1, dora , 500 points to overtake first place

(183) All-last, South seat, turn 1, dora , 500 points to overtake first place

For (184) where 234 sanshoku seems viable,  becomes useful,  less so. Still, we keep  and drop  first since  still accepts  11 tiles for tanyao. If we can't draw around , pon  and force sanshoku instead (川村晃裕, 2017, pp. 29–30).

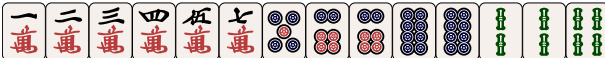
(184) All-last, South seat, turn 1, dora , 500 points to overtake first place


Hand (185) has two penchans and no completed blocks, making the 500 point difference far far away from it seems. Still, our dim hope lies in tanyao. In that regard, both  are useless. Drop  first, since, even if we draw  and  ryanmen is a furiten taatsu, we can still chi  to get rid of this taatsu. Dropping  makes  ryanmen furiten and it would be very hard to get rid of the furiten taatsu while also incorporating . Do not immediately drop  after ;  draw which gives us pairs backfires a lot (川村晃裕, 2017, pp. 53–54).

(185) All-last, turn 1, non-dealer, dora , 500 points to overtake first place


Forcing Tanyao when Menzen is Also Close

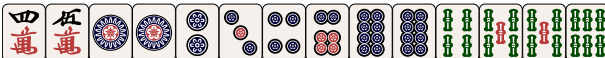
Hand (186) looks far better than previous hands and menzen tenpai is within reach. Even so, opening our hand with tanyao is still faster. In particular, in (186), there are  pairs and **2(8) comes out fairly easily**. Thus, drop  here and force tanyao (川村晃裕, 2017, pp. 41–42).

(186) All-last, turn 1, non-dealer, dora , 500 points to overtake first place


If our hand is better such as (187) which is perfect 1-away if we drop , dropping  is fine for very early turns.

(187) All-last, turn 1, non-dealer, dora , 500 points to overtake first place


Hand (188) looks like we can force tanyao by dropping  pair, but doing so will lose  8 tiles of acceptance. Notice that,  drop not only give us the widest 1-away which accepts  20 tiles, 456 sanshoku is also possible with  chi and then  drop. Therefore, drop  here (川村晃裕, 2017, pp. 43–44).

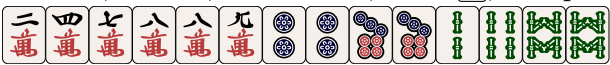
(188) All-last, turn 1, non-dealer, dora , 500 points to overtake first place


Likewise, for (189),  drop renders  useless, and it actually increases our shanten, unlike 

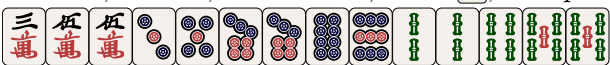
drop from (188). Widest 1-away we can get from (189) is 16 tiles. In that case, it is better to drop  and maintain 1-away with 12 tiles of acceptance, and after we draw or chi any of , we can opt to drop  entirely. This is similar to (175) (川村晃裕, 2017, pp. 43–44).

(189) All-last, turn 1, non-dealer, dora , 500 points to overtake first place

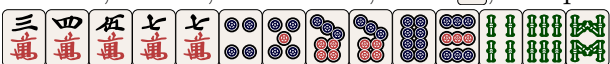

Hand (190) has 4 pairs, and chiitoe seems closer than normal hand, but in agaritoppu, tanyao should still be the first priority. Still, to make chiitoe our backup plan,  drop then becomes the best course of action. If we want to keep the possibilities of a normal hand and a chiitoe hand and drop , our hand becomes way slower (川村晃裕, 2017, pp. 57–58).

(190) All-last, turn 1, non-dealer, dora , 500 points to overtake first place


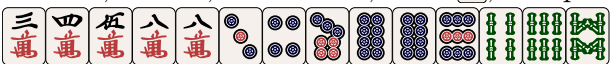
Hand (191) is mostly tanyao tiles and forcing tanyao by dropping  seems fine. However, the pairs of this hand are all middle tiles, so they would come out harder than usual. Thus, for (191), drop  and keep menzen possibility while also not overlooking 345 sanshoku chance (川村晃裕, 2017, pp. 59–60).

(191) All-last, turn 1, non-dealer, dora , 500 points to overtake first place


Hand (192) too has middle tile pairs, and even if we drop , the acceptance of  overlaps with that of . It is better to simple take the 1-away with  drop (川村晃裕, 2017, pp. 59–60).

(192) All-last, turn 1, non-dealer, dora , 500 points to overtake first place


If we change the hand a little bit and make it (193) which now has pairs of 2(8) tiles, we can force tanyao with it. Drop  (川村晃裕, 2017, pp. 59–60).

(193) All-last, turn 1, non-dealer, dora , 500 points to overtake first place


Keeping Tanyao as Plan B

Before turn 7 or until someone opens their hand, we can keep tanyao possibility in our hand instead of safe tile. This particularly works well for cases where we only have one tanyao bottleneck such as  ryanmen. Scoring tanyao not only increases our hand value, in cases where we are agaritoppu or we have dora 3 it would also increase our win rate by allowing us to open our hand (川村晃裕, 2017, pp. 155–158).

When we look at hands like (194), we should not only think that  draws give us perfect 1-away, but also keep in mind that any of  gives us tanyao chance. This tanyao chance is particularly helpful when we are agaritoppu or we have dora 3. Suppose that we draw  and our hand becomes (195), we can chi  and drop . Even with  draw as in (196), we can chi  and drop .

(194)

(195)

(196)

If our hand has no dora, the increase in value (1000 or 1300) brought by tanyao does not justify the loss in defense. Thus, for (197), we keep  and tsumogiri³² anything other than .

(197) Dora 

																			
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Likewise, hand (198) has a bottleneck , so we keep  here in hope for tanyao and sanshoku. For hand (199), however, since its tanyao is already confirmed, and there is already dora 2, there is no need for sanshoku, and  becomes useless. Drop  in (198) and  in (199).

(198)  Dora

(199) Dora 

Previously we learned that, in the context of 6 blocks, drop  from the skipping pairs  is usually better especially when there is tanyao chance. Even when we compare skipping pairs to floaters, we still must always keep tanyao in mind. For hands from (200) to (202), we are comparing  to a floater . Even though our hand already has 6 blocks and  does not contribute to acceptance, with this particular shape , we can opt to chi  and drop , so it contributes to hand value and speed.

If the hand has 3900 or 3900 potential after we open it, drop  (川村晃裕, 2017, pp. 165–170). Hand (200) has dora 2 and is already 3900 if we open our hand, and drawing any of the aka doras gives us 7700. Hand (201) only has dora 1, but its   ryanmen accepts dora  and  accepts aka dora , so it still has 3900 potential. Hand (202), however, only has one dora and

³²*Tsumogiri* refers to the action of drawing a tile and immediately discarding it. The word is a compound of *tsumo* “draw” and *giri*, derived from *kiru* “to cut.”

 is useless, so this hand has no 3900 potential and is thus not fit for calls, and we drop ad.

Dora 

Dora 

Dora 

and (203) looks like pinfu hand. Except for  which give us tenpai or perfect ay or headless 1-away with anko, keep in mind that  draws allow us to pon. In flat situation,  is useless, but it is crucial in all-last agaritoppu situations (or dora 3) (川村晃裕, pp. 17–18).

All-last, non-dealer, haipai, dora , 500 points to overtake first place

Deciding between Tanyao and Dabuton

If **menzen tenpai is close, drop dabuton** (福地誠, 2017, pp. 138–139). If we drop 東東 from (204), we still have enough good shapes for tenpai. Dabuton and dora 1 is 5800, but if we stay closed, with riichi tanyao dora 1 we score 7700, and with ippatsu, tsumo, or uradora we can reach 11600. Therefore, value wise, dropping 東東 and go for tanyao is better.

East round, East seat, turn 6, dora 

If we already opened our hand like (205), without opportunistic yakus such as ippatsu, the value of dabuton dora 1 becomes greater than tanyao dora 1. Even if we cannot reach tenpai in time and someone calls riichi, 東 is still generally safer than other tanyao tiles. We drop 𠄎 first. After that, compromise on 2900 if the ryanmens get filled first, or drop 𠄎 when we draw 福地誠 (福地誠, 2017, pp. 143–144, p. 218).

East round, East seat, turn 6, dora

Deciding between Tanyao and Yakuhai

Anything other than ryanmen ryanmen or dora 3 means we keep yakuhai pair (川村晃裕, 2017, pp. 177–180). For hands from (206) to (208), we just called  chi on turn 7 to get rid of a kanchan, and we can either drop  for tanyao or drop any of the other taatsus for chun. Some may be thinking that if  does not come out this late, *mochimochi*³³ is very likely the case, and we cannot win our hand without dropping . Paradoxically, dropping  means that it will get picked by our opponent and make it harder to win our hands.

Therefore, there is little merit of dropping . If it really is *mochimochi*, we are giving others advantage; if it is not, we lose defense when facing riichi. Only drop  pair if our taatsus are all ryanmens or we have dora 3 and we must get tanyao confirmed. Drop  from (206),  from (207), and  from (208).



2.3 Pinfu

Forcing Pinfu when Our Hand Has No Dora

Force pinfu when our hand has no dora or other yakus (土井泰昭, 2014, pp. 200–201). Hands like (209) very likely end up rii-nomi with little value, the cheapest 1300. Efficiency wise, the  that we just drew is useless. If we drop  here, considering that unseen doras are more likely end up in others hands as the game progresses, most likely we will face riichis from others that have values higher than average³⁴. From the perspective of defense and hand value, we drop  here.



(210) is a similar hand, but we have 1 han from tanyao. With 1 han in our hand, we can become more aggressive and keep maximal acceptance. Drop .

³³Derived from Japanese *motsu* “to possess,” it refers to a state where two opponents have two copies of one kind of yakuhai and thus they can never call their yakuhais.

³⁴If all  are seen, it is another story.

(210) East-1, West seat, turn 5, dora 

We must drop any of the ryanmens of (211). In a ruleset with akadoras, 2-5 or 5-8 waits are stronger than 3-6 or 4-7 waits, so we drop  here. If our only pair  is not dora, drop  first, so that when we luckily draw , we can drop  instead. If  is dora, drop  first, so that when we luckily draw , we can choose other ryanmens to drop instead (川村晃裕, 2017, pp. 13–14).

(211)

Deciding between Pinfu and Yakuhai

If we have (212) in a flat situation, since it already has good shapes, we can drop  and go for menzen riichi with pinfu. If we are agaritoppu, pinfu itself becomes less important and being able to use  becomes more important. Our instinct is to drop any of the ryanmens and keep two pairs, but we should drop  pair instead – so as to not lose the chance to directly call riichi on any draws of . After dropping  pair, any of  draws gives us perfect 1-away (川村晃裕, 2017, pp. 179–180).

(212) Dora 

2.4 Iipeiko

Since iipeiko worth only one han and is limited to menzen hands only, usually we do not go against efficiency and choose bad shapes to force iipeiko. Shapes such as  only have a fifty-fifty chance of giving us iipeiko. Shapes such as  guarantee us iipeiko when our hand is completed, but it is very hard to get this  kanchan filled since we already have one copy of its acceptance. Therefore, we keep ryanmens over , and we keep other kanchans over  when one more  is dropped (ネマタ, 2014f). In this regard, we drop  from (213) (多井隆晴・郡道美玲, 2021, pp. 121–122).

(213) North seat, turn 6, dora 



When our hand has no dora or has dora 3, we can lean a little bit towards iipeiko chance when choosing which floater to keep. Hand (214) has three floaters

Table 50: Acceptance of shapes resulting from one tile drop from various combinations of three skipping pairs and taatsus.

(215)  is once cut, dora 

Three Skipping Pairs

For shapes such as  where we have three pairs, efficiency wise, dropping the middle tile gives us the most acceptance as is shown in Table 50. However, if we keep this 6-tile shape, we have iipeiko upon  draw. Therefore, the discussion of this shape will involve how much we want iipeiko even if that means sacrificing efficiency (ZERO, 2020, p. 35; ヨーテル, 2021c; G・ウザク, 2019, pp. 114–117).

For example, the drop candidates of (216) are . Their merits are,

- Merits of  drop
 -  draws give us  shanpon wait which has high win rate
- Merits of  drop
 - Iipeiko chance on  draw
 - After done dropping this pair, we have space for safe tiles

If our only goal is to riichi, drop  and hope that our wait is  and/or . If there are one or more  visible on the table which makes our wait near dead, or we are in a point situation which favors dama, then drop  and rely on iipeiko chance (ZERO, 2020, p. 35).

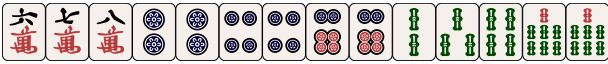
(216)

Hand (217) has its three skipping pairs closer to the middle. Unlike the possible  shanpon wait in (216), the best shanpon wait we can get from (217) is  and it already has lower win

rate than . In such cases, we lean towards dropping  (ヨーテル, 2021c).

(217) 

For hands like (218) which has tanyao confirmed, we can still either aim for iipeiko, which has theoretically the highest expected value, or drop  and hope for ryanmen improvement of . Since we have tanyao confirmed, we can call and wait for improvements even after we reach tenpai. The only ryanmen improvement of  is , while  improves on  ( cannot accept  if we force tanyao.) Therefore, in situations such as agaritoppu where hand value is not to be maximized, drop ; otherwise drop  (G・ウザク, 2019, p. 117).

(218) 

2.5 Honitsu and Chinit-su

In the following cases, we ignore general tile efficiency and force honitsu,

- Acceptance is narrow as there are many bad shapes and thus far from menzen tenpai (渋川難波, 2023c)
- Hand value is mangan (or near mangan) even after call (平澤元気, 2021a)
- Hand is still cheap (below 2 han) if we go straight for riichi (平澤元気, 2021a)
- We have two pairs of yakuhai, which we call *yakuyaku* (ZERO, 2020, pp. 96–97)
- We have one pair of yakuhai and the rest of our hand is abysmal (ZERO, 2020, pp. 97–98)

Honitsu can only use tiles from one suit and honor tiles, so, in terms of pure efficiency, forcing honitsu sacrifices efficiency. Generally, this loss in efficiency is justified by its value (2 han minimum, easily becomes mangan with dora or yakuhai) and its defense (honor tiles generally have lower deal-in rate.) When we have abysmal haipais, forcing honitsu is usually a good option, value wise and defense wise.

Forcing Honitsu in Case of Yakuyaku

Mahjong is a game of building mangan. If we have two yakuhai pairs, we can maximize our hand value and easily reach mangan by forcing honitsu (ZERO, 2020, pp. 96–97). The value of hand (219) is 1000 or 2000 if we just follow pure efficiency. If we force honitsu, it can be 3900 or 7700, way bigger than before. Drop .

(219) East-1, North seat, turn 1, dora 


Hand (220) has a completed souzu block. If we are super early in a round like before turn 4 and yakuhai comes out, we can call pon and drop the entire souzu block to force honitsu (ZERO, 2020, p. 97). This is the extent we can deny pure efficiency and maximize hand value.

East-1, North seat, haipai, dora 

Hand (221) has 6 blocks. With 4 pairs in our hand, we can reach mangan with either honitsu or toitoi. In this regard, the weakest tiles are    . From there, we have two options (独歩他, 2017a, pp. 98–103),

- Drop  for  acceptance and maximal toitoi and honitsu chance.
- Drop  if we are okay with a 2000 hand. Drop  before  because  is jikaze.

(221) East-1, West seat, turn 1, dora 

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Hand (222) is similar to (221), but its manzu part becomes a kanchan. In this case,  kanchan becomes even weaker and we can drop it without much doubt (独歩他, 2017a, pp. 98–103).

(222) East-1, West seat, turn 1, dora 

Think Twice When There is No Yakuhai Pair

When there are no yakuhai pairs in our hand, forcing honitsu will mostly give us 2000 or 3900 with one dora. This holds as well for cases where we have otakaze pairs. Therefore, we are less incentivized to force honitsu for offense. For defense purposes, though, honitsu still remains an option.

We already have a head and iipeiko in (223). Having iipeiko already confirmed means that if we reach riichi this hand we already have 2 han. Taking the possibility of tanyao and pinfu into consideration, staying menzen has more value than forcing honitsu and opening our hand, especially when there are no yakuhai pairs and the only thing that could increase our value is aka dora. Still, there is a slim chance of chinitsu. Therefore, we drop  first and keep  (渋川難波, 2023c).

(223) South-1, South seat, turn 1, dora , last place

Hand (224) has 5 lone honor tiles. Drawing tiles near floating number tiles is generally faster than drawing duplicate copies of honor tiles. In addition, there are no head confirmed in our hand; even if we draw duplicates of honor tiles, calling pon on them does not speed our hand up. Therefore, drop honor tiles; drop yakuhais first, in particular, since otakaze pairs can still contribute to pinfu (ゆうせー, 2020c).

(224) East-1, West seat, turn 1, dora 


Hand (225) and (226) are both hands with narrow acceptance and mostly consisting of bad shapes of non-dominant suit. While it is true that we do not have yakuhai pairs and increasing our hand value is difficult, defense wise the otakaze pairs provide us a lot of safety. Therefore, we force honitsu for defense purpose here (渋川難波, 2023c).

(225) East-3, East seat, turn 2, dora ,  is once cut


(226) East-3, East seat, turn 2, dora ,  is once cut


Hand (227) has no yakuhai pair at all. Still, this is agaritoppu, and we must force some yaku to take first place. For this kind of messed up hand, honitsu becomes our hope, with chanta also slightly probably. Therefore, drop , and prepare to call everything possible (川村晃裕, 2017, pp. 37–38).

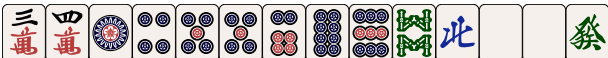
(227) All-last, West seat, turn 1, dora , 500 points to overtake first place


When There is Only One Yakuhai Pair

Having only one yakuhai pair means that we are in the middle ground between the sweetspot of honitsu value increase and menzen riichi. Deciding whether to force honitsu or not will largely depend on how good the rest of our hand is.

With one yakuhai pair and two lone yakuhai tiles, we are in a state referred to as “ready for yakuyaku.” In this state, we can tolerate dropping floating 1289 tiles of lesser suits (ZERO, 2020, p. 97). In this line of thought, we drop  from (228) and  from (229).

(228) East-1, North seat, turn 1, dora 


(229) East-1, North seat, turn 1, dora 


Likewise, we drop  from (230) and get prepared for pinzu honitsu (平澤元気, 2021a).

(230)

difficult to make more blocks in manzu suit. Drop 

(231)

tiles, but the shapes in lesser suits are all ryanmen.

(232)

pivot to toitoi. Considering also the slim chance of

(233)

Drop The Suit with Second Most Tiles First

pinzu tiles first, and delay the timing when opponer

(234)

difference in the number of pinzu and souzu tiles.

(235)

Actively seek ways to make yakuyaku mangan (ヨーテル, 2023g). As mentioned previously, yakuyaku gives us 2 hans that is the sweet spot where, for 2 additional hans, our value increases from 2000 to 7700, a whopping 5700 points of difference. Honitsu is the most common way to achieve mangan, but keep in mind that toitoi is also an option.

(236) 

to call than outside tiles, so we drop  here first.

(237) 

With one dora in our hand, the increase in hand value now becomes limited. Suppose that we have all yakuhai poned, our hand value is already 3900 minimum, and forcing honitsu will only make it 8000, since its value is capped at mangan. Therefore, we do not force 2 han yakus, and for hands like (238) where we also have a ryanmen in lesser suits, it is perfectly fine to proceed with 3900. Drop .

(238) 

and chanta which are 1 han yakus that brings us mangan. Drop 

(239) 

none of our yakuhai pairs are ponned, which means that, when we reach tenpai, we either have

one pair as our head, or wait on yakuhai shanpon, and our hand value is still 2000 minimum. Therefore, we force honitsu here and drop .

(240) East-1, South seat, turn 7, dora 


Hand (241) likewise is a hand where we did not get to pon our yakuhai pairs in time. Drop  and force toitoi for mangan.

(241) East-1, South seat, turn 7, dora 


Cases Where We Advance Yakuyaku Without Forcing Yakus

When our hand is slow, the increase in value from forcing 2 han yaku outweighs loss in speed, but this is not the case when our hand already has enough good shapes like (242). Drop  and aim for a fast 2000 to get to next round.

(242) East-1, South seat, turn 1, dora 


If we are in huge lead and just want to move to the next round, or we are agaritoppu, forcing honitsu or toitoi is also not a good option.

2.7 Sanshoku Doujun

Asada Tetsuya once said, “look for sanshoku³⁵ as soon as you see haipai.” In the early days of mahjong when riichi (in the sense that we declare our tenpai in the middle of a round), ippatsu, uradora and stuff didn’t exist, building your hand with yakus such as sanshoku was important. As riichi and uradora became widespread, the importance of sanshoku dropped, and it is not as popular nowadays. Some would even go so far as to say that sanshoku is a “luck” yaku.

Still, being a 2 han yaku (1 han if our hand is open) and a yaku that only uses 3 blocks, sanshoku is still a useful yaku. The following criteria will help us decide when to force sanshoku (ネマタ, 2014g):

- How easy it is to get sanshoku confirmed.
- Loss of acceptance in the case when it is hard to confirm and we have to force it.
- The merit of increase in value with 2 hans of sanshoku or just the yaku itself.

³⁵There is another yaku called *sanshoku doukou* which requires three triples with the same number, i.e., , , and . Since sanshoku doukou is far less common than sanshoku doujun, in most cases, *sanshoku* refers to sanshoku doujun.

When our hand has no dora, building value to make it worth fighting becomes crucial. When our hand has a lot of dora but also far away from tenpai, having a yaku to back up our calling is also important. In these two cases, we tend to force sanshoku.

If our hand is close to tenpai, or our head is not confirmed, sanshoku becomes harder to score. Hand (243) already has 7 tiles necessary for 123 sanshoku, but there is no confirmed head. The most likely outcome is drawing some good shape around yonrenkei and drop to confirm as our head. Going for pinfu is more efficient (ZERO, 2020, p. 67).

(243) South seat, turn 5, dora

(248) South seat, turn 3, 北



If our hand has only one dora, there is no big difference in terms of expected return. Adjust your judgement according to actual table situation.

Go Big or Go Home with Bad Haipai

Hand (249) is 4-away and has only one good shape. If our hand is messed up anyway, we can try to force yakus to increase value or move on with quick hands. From this hand, we can see yakuhai and tanyao. Note that with 5 tiles of 123 sanshoku already in our hand, sanshoku is also viable, especially when we do not have enough blocks. Therefore, drop the most useless tile . Even though it is in a  block, notice that it is not part of any of the yakus mentioned previously (ZERO, 2020, p. 64).

(249) South seat, turn 1

Keep Seeds of Sanshoku Over Bad Blocks

Hand (250) has 6 blocks. According to 6 block theory, we drop floater  and take maximal acceptance of 22 tiles. However, dropping  and making lower³⁶ pinzu part one block only lose 2 tiles of acceptance, and we get to keep the “seed” of 789 sanshoku. Therefore dropping  is good (ZERO, 2020, p. 64). Another way to look at this is to treat  as a block, so we are still maintaining 6 blocks in our hand, and we are ready to drop  when we are closer to tenpai.

South seat, turn 3, dora 

(250) 

Likewise, in the case of (251), it would be very difficult to make two mentsus from , so we drop  and keep  with allows us to aim for 789 sanshoku (ネマタ, 2014g).

(251) 

6 Block Theory Sanshoku Edition

If you have read 1.3, you will notice that dropping  from (252) will give us maximal acceptance as well as guaranteed pinfu. Notice that we also have 7 tiles required for 789 sanshoku as well, but since our shapes in upper parts of suits are ryanmens, we could easily end up not getting sanshoku, and keeping  makes us lose 3 less tiles of acceptance than  drop. Therefore, we will keep  only when we have no dora and have to chase for value; if we have at least one dora, we should drop  (ネマタ, 2014g).

³⁶The “upper” part of a suit refers to tiles with greater number and the “lower” part refers to tiles with smaller number.

[illegible]

(258)

Sanshoku with a floater as its middle tile is weak (ZERO, 2020, pp. 65–66). Take (259) for example. can be the seed of lower part sanshoku and is dora, so we cannot drop either. Thus, it becomes a decision between and , which is equivalent to choosing between 123 and 234 sanshoku. As shown in Table 51, the greatest difference is draw which, if we dropped previously, gives us no yaku at all. Dropping always gives us a yaku, so we drop and aim for 123 sanshoku. We thus conclude that, with floater, 234 sanshoku is weaker than 123 sanshoku.

(259)

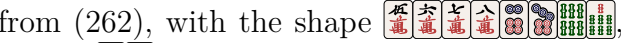
Table 51: Comparing  drop and  drop from (259). Sanshoku in parentheses means that it is not confirmed due to ryanmen waits.

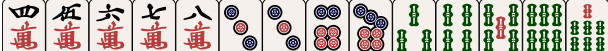
(260) South seat, turn 3, dora 

(261) 

Ryoutenbin

Ryoutenbin is a Japanese term which generally translates to “sitting on the fence,” and it refers, in the context of mahjong, to a situation where, with 13 tiles, there are possibilities of two yakus that cannot coexist. Which yaku we eventually score depends on the subsequent draws. There are ryoutenbins of sanshoku and ittsu and also ryoutenbins of different sanshokus.

If we drop  from (262), with the shape , drawing  gives us 567 sanshoku chance, and drawing  gives us 678 sanshoku chance. This shape is therefore the ryoutenbin of 567 and 678 sanshoku. While it is true that dropping  gives us sanmenchan and maximal acceptance, in a flat situation, we can sacrifice 3 tiles of acceptance and take sanshoku ryoutenbin (Chiba, 2020).

(262) 

Hand (263) becomes ryoutenbin of 123 and 234 sanshoku if we drop . While it indeed is 4 tiles less of acceptance than  drop which gives us perfect 1-away, our sanshoku chance also increases, which gives us way higher expected value than  drop³⁸. Drop  (クリアレイン, 2023b).

Turn 6

(263) 

Hand (264) also becomes ryoutenbin of 345 and 456 sanshoku if we drop . In this case, we do not have pinfu, thus there is a higher chance of undesirable rii-nomi, which further justifies loss in 4 tiles of acceptance in exchange of increased sanshoku chance (ネマタ, 2014g).

(264) 

Hand (265) also becomes ryoutenbin of 567 and 678 sanshoku if we drop any of . However, since this is a headless hand with anko, dropping  actually loses 11 tiles of acceptance, a wide gap in acceptance that do not justify increased sanshoku chance. Considering that it is more likely for 8 of 678 sanshoku to come out and complete our hand, we drop  here (クリアレイン, 2023b).

Turn 6

(265) 

If we drop  from (266) for 567 and 678 sanshoku ryoutenbin, we lose  which gives us mentanpin dora 1 mangan tenpai. In previous cases where we would trade 4 tiles of acceptance for sanshoku chance, those 4 tiles do not contribute to value, but this  puts us from 3900 to 7700, an increase too great to lose. Therefore, drop , since 567 sanshoku is more likelier than 678 sanshoku with  block (ネマタ, 2014g).

³⁸The difference in expected value justifies loss of acceptance up to dora 2. When we have dora 3, the difference in expected value is very small and we should go for maximal acceptance.

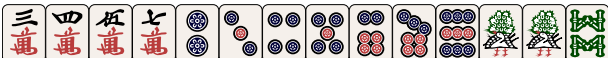
(266) 

2.8 Ikkitsuukan

Ikkitsuukan in Japanese means “to strike through in one breath,” and refers to a yaku we score when we have all nine tiles of a suit. It is often shortened to *ittsu*, or translated into English as *pure straight*. Ittsu has the same value as *sanshoku*, which is 2 han for *menzen* and 1 han for open hands. Since ittsu requires all nine tiles from a suit, there are only 3 possible patterns of ittsu (1–9 tiles in *manzu*, *souzu*, and *pinzu*), less than *sanshoku* (123, 234, 345, 456, 567, 678, 789 tiles), it is a less common yaku.

When Ittsu is Viable

We can go for ittsu if our hand has no dora, it is still far away from *tenpai*, and we have at least 5 tiles in a suit. In such case, we keep floaters in the dominant suit and drop other floaters, even if they have more acceptance. Take (267) for example where we are choosing one from . Acceptance wise,  is the weakest, but this hand has no *tanyao* no *pinfu* no dora whatsoever. Thus, keep  for acceptance of  dora, and keep  for a slim chance of *pinzu ittsu*. Drop  which is the one that contributes the least value (ZERO, 2020, p. 68).

(267) Dora 


We must drop either of  from (268). *Manzu ittsu* is further than (268) with only 5 *manzu* tiles, but it is still slightly better than  which gives us nothing but the chance of a *penchan* or a *kanchan*. Likewise, drop  from (269) (ZERO, 2020, p. 69).

(268) Dora 


(269) 

When comparing  *penchan* and  *kanchan* in (270), efficiency wise, it is better to drop *penchan* and wait for *ryanmen* improvement of *kanchan*. However, dropping  means that we lose ittsu chance on  draws. Therefore, we drop  (ZERO, 2020, pp. 68–69).

No dora
 (270) 

Dropping *ryanmen* in favor of *penchan* may sound counterintuitive, but for cases such as (271) where we have no dora and must force ittsu, two *penchans* that confirm ittsu is better than a *penchan* and a *ryanmen* that give us only *rii-nomi*. Therefore, drop  (ネマタ, 2014h). If our hand has 1 dora, follow pure efficiency and drop any of the *penchans*.

一萬 二萬 四萬 五萬 六萬 八萬 九萬

Cases Where Ittsu is Not Viable

We rarely go for ittsu when we have 1 dora, or we can have tanyao confirm if we remove non-tanyao blocks, even if we have 7 tiles in a suit. While it is true that ittsu has more value than tanyao when stay closed, the shapes in the ittsu suit are fixed and cannot upgrade to ryanmens. If we have at least 1 han, this inflexibility becomes undesirable.

Hand (272) has 7 distinct manzu tiles, but dropping  will not make us lose the acceptance of  , and  allows us to drop  and confirm tanyao. It is thus the right choice to drop . (ZERO, 2020, p. 69).

(272)

Likewise, if there is 1 dora in (273), drop  and go straight forward. Only drop  ryankanchan when there is no dora (ZERO, 2020, p. 70).

(273) 

Dropping $\begin{array}{|c|c|}\hline \text{一} & \text{允} \\ \hline \text{通} & \text{通} \\ \hline\end{array}$ from (274) gives us tanyao, so we do not need ittso here. Drop $\begin{array}{|c|c|}\hline \text{一} & \text{允} \\ \hline \text{通} & \text{通} \\ \hline\end{array}$ (ZERO, 2020, p. 70).

(274)

Hand (275) too does not need ittsu if dropping 一萬二萬 already gives us pinfu (ネマタ, 2014h).

[illegible]

Golden 1-Away

Golden 1-away is a subset of kutttsuki iishanten that is also a ryoutenbin of sanshoku and ittsu. Take (276) for example. If we drop ,  draws give us pinzu ittsu chance while   draws give us 789 sanshoku chance. Taking the iishanten with ittsu and sanshoku chances has higher expected value than  drop which gives us only pinfu, so we drop  here (Chiba, 2020).

(276) 

Hand (277) too is golden 1-away. Golden 1-away does not always come in shapes like (276) where there is clearly a floating . For the combination of yonrenkei and ryanmen-toitsu, (277) is also a kuttsuki iishanten.

(277)

Hand (278) becomes golden 1-away if we drop . Dropping gives us kuttsuki iishanten with two yonrenkei shapes, whose 53 tiles of acceptance is undeniably the most, but its bad wait rii-nomi draws is also the most (渋谷難波, 2024a). Since ryanmen implies 345 sanshoku chance, and we need pinzu yonrenkei for ittsum, becomes the most valueless tile.

(278) East-1, East seat, turn 5, dora

Hand (279) becomes golden 1-away if we drop . However, dropping is not too bad either, since it still gives us a wide acceptance of mentanpin (unless with unfortunate draws.) Therefore, drop for the best chance of good wait tenpai (ネマタ, 2014h).

(279)

However, if the pair becomes and tanyao is gone, drop for value³⁹.

(280)

Choosing between Sanshoku and Ittsu

Since both sanshoku and ittsum uses 3 blocks, we cannot have both when we are dropping one block from a hand with 6 blocks. In most cases, **keeping sanshoku is better** (ネマタ, 2014h) since sanshoku can coexist with tanyao.

Hand (281) looks like 789 sanshoku or pinzu ittsum. However, it has 6 blocks, and we must drop one block from any of , , and . To complete ittsum, we need exactly , which is one more kind of tiles than 789 sanshoku and more difficult to score. Therefore, we drop ittsum here and drop . If we draw and drop , we can even have 678 sanshoku and tanyao at best.

(281) East round, North seat, turn 7, dora

Hand (282) already has pinzu ittsum confirmed. If we drop any of the ryanmens and have the other one filled before does, we end up 5200. On the other hand, if we drop , we have 123 and 234 sanshoku ryoutenbin, and draws give us tanyao chance. Therefore, we do not go for ittsum here. Drop .

(282) East round, West seat, turn 8, dora

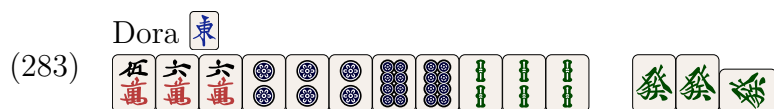
³⁹The result I got from simulator.

2.9 Toitoiho

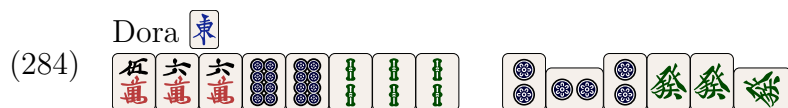
Toitoiho, or *toittoi* for short, has 2 hans regardless of hand being open or not, making it highly compatible with other yakus such as yakuhai and easily giving us high value. However, since *toittoi* hands mostly consist of pairs and triplets, when our opponents call *riichi*, we have less kinds of tiles and thus less possible safe tiles (ヨーテル, 2023d). In this section, we will look at cases and see if it is worth sacrificing tile efficiency for value; for cases where others drop tiles and we are presented the option to open hand and go for *toittoi*, see Section 3.

From 1300 to 5200

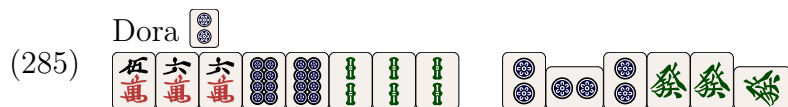
If we drop  from (283), its value is just 1300. If we force *toittoi*, we have half the tenpai tiles but 4 times the value, a good trade-off. Drop  (ZERO, 2020, p. 57).



Even if we have two open sets and we are more exposed to the danger of *riichi*, we can still force *toittoi*. Drop  from (284) (ZERO, 2020, p. 57).



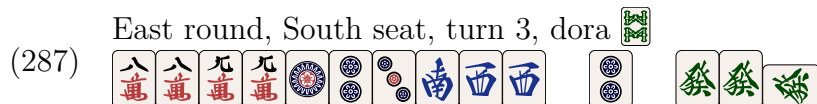
If we already have *mangan* with *ryanmen* tenpai, there is no need to force *toittoi* for *haneman*. If we halve our tenpai tiles, our value only becomes 1.5 times higher. Drop  (ZERO, 2020, p. 57).



Hand (286) too has 4 pairs and we can go for *toittoi*. Drop  (ZERO, 2020, p. 57).



Hand (287) is the moment when we draw . Seeing  already completed, people would drop  without thinking too much. Do keep in mind that, if we keep , there is a chance for *toittoi* and 5200 points. If it is before turn 4, drop  and go get that 5200. If it is later than that, drop , and keep *toittoi* route as our backup plan (ZERO, 2020, p. 58).



Dora 

Ryanmen 5200 is Sufficient

(289) 

(290) 

Forcing Toitoi

- Forcing toitoi gives us 5200 or higher value
- There are 4 pairs and there is at most 1 middle tile pair among them

Dora [illegible]

⁴⁰*Sotogawa* in Japanese means “outside,” and it refers to the outside of early discards in mahjong context. *Sotogawa* is generally considered safe because early discards are often not connected or related to other shapes in our hand, and cases where we drop  from  early are rare (but not impossible.) The act of deliberately dropping  from  early is called *sakigiri*.

(292)

九真

(293)

Cases Where We Do Not Go for Toitoti

in

(294)

2.10 Chiitoitsu

la

From Three Pairs to Chiitoi

C.

(295)

W

the weakest kanchan  (ZERO, 2020, p. 83).

(296) East round, North seat, turn 3, dora

Forget about Chiitoitsu When Our Hand is Full of Good Shapes

For both (297) and (298), we *could* drop any of the tiles with a single copy and make our hand ryoutenbin of chiitoitsu 1-away and normal hand 2-away, but we *should not* do that with (297) which has 3 ryanmen-toitsus. With  anko, we can open our hand and proceed just fine; we are 2-away from normal hand indeed, but it is faster than chiitoi. On the other hand, for (298) which has one kanchan-toitsu, dropping  loses  acceptance but gains  acceptance, so it is actually 2 more tiles of acceptance. Therefore, we confirm a ryanmen in (297) and drop  from (298) (ZERO, 2020, p. 83).

(297) East seat, turn 5, dora 

(298) East seat, turn 5, dora 

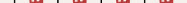


Hand (299) has neighboring pairs and our acceptance is less ideal, but we still exclude chiitōi from the possibilities early in a round and drop . When we start discarding second row discards, drop   and advance normal hand and chiitōi in parallel (ZERO, 2020, p. 84).

(299) Dora 

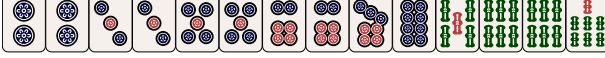
三萬 三萬 四萬 四萬 七萬 八萬 11 11 11 11 11 11 中 中 中

Hand (300) has a kanchan and is much less ideal than previous hands. No matter how early it is, drop  and advance normal hand and chiitai in parallel (ZERO, 2020, p. 84).

(300) Dora  

As shown in Table 52, dropping either of  from (301) will give us the same number of 17 tiles of acceptance. If we drop , the neighboring pairs  is forced to serve as head candidates and  draws cannot make it mentsu + ryanmen. Still,  draw cancels tanyao out, so it doesn't hurt much when it backfires; the draws around  would hurt more than that. Therefore, we drop  here; if we happen to reach chiitai tenpai, take it (G・ウザク, 2016, p. 18).

(301) East-1, East seat, turn 5, dora 

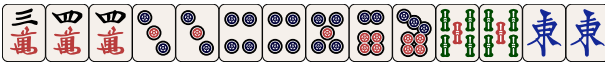


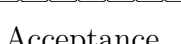
Drop	Acceptance
	 17 tiles
	 17 tiles

Table 52: Acceptance of (301).

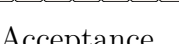
Hand (302) has enticing dora dabuton. However, we are already at turn 9, and the later it is the harder it is for dora to come out. Thus, we would take chiitoe as our plan B. Dropping either of  gives us ryoutenbin of normal hand of chiitoitsu, and the difference lies in the shapes after we call  pon, as shown in Table 53. If we keep  mentsu, it would be difficult for  to function as ryanmens; conversely, and somewhat unexpectedly, dropping  in tandem leaves us  ryanmen +  pair +  ryanmen-toitsu. Therefore, drop , which gives us chiitoe possibility and the best acceptance after  pon (G・ウザク, 2016, p. 20).

(302) East-1, East seat, turn 9, dora 



Drop	Acceptance
	 17 tiles
	 17 tiles
	 17 tiles
	 17 tiles

(a) With  already dropped.

Drop	Acceptance
	 19 tiles
	 19 tiles
	 19 tiles

(b) With  already dropped.

Table 53: Shapes of (302) after  pon.

Setting Up First Row Discards

Efficiency wise, it should be  drop from (303). However, this hand is very likely rii-nomi, so we would like to have chiitoe as our backup plan. We can drop either of  to keep acceptance in souzu suit, or we can drop , and make  look like sotogawa, which increases our win rate (ZERO, 2020, pp. 84–85).

(303) South seat, turn 4, dora 



Likewise, drop  first from (304), even though both  are equally bad for a chiitoe tanki wait.

(304) East seat, turn 4, dora 


Note that in both cases we are still at turn 4 where our discards are very likely considered sotogawa. If it is already second row discards, sotogawa trap becomes ineffective, and we can stop thinking about how our river looks.

Passive Chiitai

Chiitai hands, much like toitai hands, lack defense. When we are in the middle of a round where it is likely that our opponents would call riichi next turn, the floater we keep in our hand would be more defense oriented. Therefore, while  is twice cut and gives us furiten if we end up waiting on it, we should still drop  so that, even if some one calls riichi, we can still drop safe tiles while maintaining chiitai 1-away (ZERO, 2020, p. 85).

(305) East seat, turn 10, dora ,  is twice cut and furiten


Choosing between Chiitai and Toitai

According to ネマタ (2014i),

- The following conditions favor toitai,
 - We have an anko.
 - Yakuhai or honitsu is feasible.
 - We have terminal pairs, or pairs close to terminals.
 - Agaritoppu in all last
- The following conditions favor chitai,
 - We have a pair which is already twice cut
 - We have no anko
 - We have no other yaku
 - We have a lone dora
 - Our pairs are middle tiles
 - We are in the lead and would like to avoid unnecessary risk

Hand (306) looks like  drop for  pon or  drop for pinfu.  drop would only leave us a 1000 hand, so unless we are agaritoppu, it would not be ideal.  drop is the menzen way, but it could also be slow while none of the menzen yakus such as pinfu and iipeiko are guaranteed. In this regard, the possibilities of chiitai and toitai would give us maximal expected value.

For chitai and toitai,  would be equally useless, but since we still have  pair, to not completely confine ourselves to chitai, consider the shapes after we actually call  pon,  would counterintuitively be the best drop here (ネマタ, 2014i).

(306) Dora 北

三萬	三萬	四萬	七萬	七萬	八萬	八萬	九萬	二風	二風	三風	三風	中	中
----	----	----	----	----	----	----	----	----	----	----	----	---	---

Hand (307) has both conditions that favor toitoi and conditions that favor chitoi. Considering that, in the case we end up chitoi tanki, our win rate plummets, so we drop 三風 and go for toitoi unless we are in the lead (ネマタ, 2014i).

(307) Dora 北

三萬	三萬	三萬	八萬	八萬	二風	二風	三風	三風	三風	北		發	發
----	----	----	----	----	----	----	----	----	----	---	--	---	---

- Conditions that favor toitoi:
 - We have an anko
 - We have yakuhai pair 發
 - We only have one middle tile pair 三風
- Conditions that favor chitoi:
 - We have a lone dora

For hands with two anko and four pairs, our choice will be based on (1) if there is yakuhai pair and (2) how many ideal chitoi tanki wait tiles are left. If we drop 三風 from (308), any draws other than 三風二風 would give us instant chiitoi tenpai; on the other hand, if we drop 三風, we can either ponten⁴¹ or even go for suuanko. Since we already have 發發 pair, toitoi would be the best course.

(308) Dora 北

三萬	三萬	三萬	八萬	八萬	二風	二風	三風	三風	三風	三風	發	發
----	----	----	----	----	----	----	----	----	----	----	---	---

3 Call Judgement

Calls, melds, or *fuuro*, refer to *chi*, *pon*, and *kan*, the methods you claim a tile from other players discards. When we call and open our hands, the state of *menzen* expires, and *menzen yakus* such as *riichi iipeiko* and *pinfu* also become impossible, thus decreasing the expected value of our hand. We cannot touch the open part of our hand, so we will defend against *riichi* with fewer tiles as well. The merits of calling are thus speed and confirmation of *yaku*, as summarized in Table 54.

	Open	Menzen
Speed	○	×
Value	×	○
Defense	×	○

Table 54: Comparison between open hands and *menzen* hands (平澤元氣, 2019, p. 87).

Call judgement is based on the following factors:

- How fast we can reach tenpai by opening our hand

⁴¹A portmanteau of *pon* and *tenpai*, meaning that we reach *tenpai* after we call *pon* and drop a tile.

- Does the hand “look like menzen”?
- Can we get rid of bad shapes with calls?
- How good and how far from tenpai is the result shape?
- Are there any other fast opponents?
- How much defense there is after shortening our hand
- How much the value of our hand decreases (except for cases such as agaritoppu where value is relevant)
- Whether the open part would alarm other players (and whether they care)
- Turn; the later it is, the better it would be for us to open our hand for tenpai

It is very difficult for humans to perform real-time calculations and weigh these factors, and the decision process can be laden with preferences and biases and easily affected by many other things even outside of mahjong table, so often different people have different call judgement. This section is compiled with material from various sources and put in a way as coherent as possible, but there still could be some disagreements between these sources. We could have some leeway in call judgement in very specific conditions, but the general guidelines should stay the same.

As a rule of thumb, if any of the following conditions are met, go ahead and make the call (平澤元氣, 2019, pp. 36–41, p. 45, pp. 58–61):

- The increase in value brought by riichi is limited. That is, there are no menzen yakus such as iipeiko and pinfu.
- The value of our hand is already mangan.
- We can get rid of bad shape or drained up ryanmens.

If all of the following conditions are met, do not call (平澤元氣, 2019, p. 48; Chiba, 2020, p. 216):

- Cheap; if we open our hand, its value is no more than 2000.
- Far away; there is no head, or it has so many bad shapes that it is very likely we have to make 3 calls.
- There are no safe tiles.

3.1 Tanyao

When we open our hand with tanyao as its main yaku, it is called *kuitan*. Since *kuitan* is completely made of middle tiles, while it is very easy to score, it also comes with a risk of low defense. For players with menzen preference, the criteria for *kuitan* would be higher, and cheap *kuitan* are rarely forced except for situations such as agaritoppu. As a rule of thumb, the criteria for *kuitan* are (ヨーテル, 2024a; 独歩他, 2017a, pp. 170–175),

- Ponten or chiten for hands worth 1000 points
- Call all bad waits for 3900 hands; leave ryanmen calls for mid round.

Only Call for Tenpai for Hands Worth 1000 Points

Except for agaritoppu, regardless of the shapes of the rest of our hand, regardless of the callable part is bad shape or not, refrain from early cheap calls (ヨーテル, 2024a; 独歩他, 2017a, pp. 170–

175; 平澤元氣, 2019, pp. 127–128). Therefore, do not open (309) or (310) or (311).

(309) East-1, non-dealer, turn 4, dora 


(310) West seat, turn 4, dora 


(311) East-1, West seat, turn 5, dora 


If we can get rid of bad shape and also get tenpai, it is fine to call. We can call  from (312) (ヨーテル, 2024a).

(312) East-1, non-dealer, turn 4, dora 


Hands Worth 2000 are Tricky

Generally we can treat hands worth 2000 as if they are worth 1000 only (独歩他, 2017a, p. 179). That is, do not open unless opening our hand gives us tenpai. This 1000 points of gap is not too much of a difference.

For hands with one dora, we are at the sweetspot of reaching mangan with riichi, so we would still be inclined to stay menzen even at 1-away. According to 平澤元氣 (2019, p. 33, p. 38),

- If we have other menzen yakus, stay menzen until the end of second row discards.
- If there are no other menzen yakus and there is no much difference between menzen and open in terms of value, open our hand as soon as we enter second row discards.

Hand (313) is perfect 1-away and has pinfu chance which could make it 7700, or 5200 if there is no pinfu anyway. If we open our hand here, it will only be 2000 or 3900 if we draw any of the remaining aka doras. Thus we should not open this hand until about turn 12, unless we are agaritoppu, there are opponents who already openend their hands, or the ryanmens are drained up.

(313) East-1, West seat, dora 


If our hand is ryanmen + kanchan 1-away, we can call kanchan part regardless of turn, and leave ryanmen part after about turn 9 (平澤元氣, 2019, p. 40). For (314), we can chi  at any turn, or chi  when its turn 9 or later.

(314)

(315)

Call all bad shapes for hands worth 3900 (独歩他, 2017a, pp. 176–181). For hands like (316) which is laden with bad shapes or hands like (317) which seems close to menzen riichi, is a must call. For (316), we can also call chi. We can also pon to make our hand mangan.

(316)

(317)

(318)

If our hand is already mangan (or 7700), this merit diminishes, and in a table where your opponents would try to sabotage your visibly expensive hands, the increase in value no longer justifies the decrease in win rate. For hands such as (319) which is already full of bad shapes, a wider acceptance

West seat, turn 4, dora 

West seat, turn 4, dora 

Calls that Save Us from Bad Wait and Dropping Dora

Dora 此

Dora Dora 

East-1, West seat, dora 

89

Tanyao Calls in All-Last

When we are in all-last and there is only 500 points left for us to overtake first place or get out of last place, tanyao calls become immensely important. While it is true that each call sacrifices one turn of tsumo, our hand is no longer confined to menzen and is thus faster than menzen. There is no need to consider defense as well since it is better to fight for a slim chance for a better placement than to fold and completely give up the chance of winning. The following hands assume that we absolutely to win 500 points in all-last for a better placement unless otherwise stated.

Chi with and drop from (325). Likewise, chi with and drop from (326). Both hands are perfect 1-away, and we stay menzen in flat situations. In all-last and in dire need to win any hand, value becomes irrelevant and we'd rather get to fastest tenpai with any yaku than wait for menzen riichi which is slow and has lower win-rate. Thus, the point here is to find ways to get rid of non-tanyao tiles, or to slide completed blocks such as for tanyao (マーク 2, 2023).

(325) All-last, dora

(326) All-last, dora

Keep in mind that dropping after chi is *kuikae* which is forbidden in most rulesets. The reason why (325) and (326) are special is because is connected to other completed blocks. Do better see this, we consider shapes minus the terminal tile . The shape can be either + or + . Likewise, can be + . However, if we chi with from and drop we are left a furiten ryanmen-toitsu instead of a completed block, so this is not a call that really advances our hand. Table 55 lists all possible shapes and calls to allow us pivot to tanyao.

Shape	Combination	Call
	+	chi
	+	chi
	+ +	pon
	+	chi

Table 55: Calls that allow us to swap blocks without committing *kuikae* (川村晃裕, 2017, p. 20). In all cases, we can drop after.

There are indeed cases where we only have an isolated block and must sacrifice one safe tile for tanyao like (327). Chi with and (マーク 2, 2023).

(327) All-last, dora

Always call shimocha's discards and be cautious about calling kamicha's discards (川村晃裕, 2017, p. 19–20). If we call shimocha's discards, we will only let shimocha draw one more turn. If we call kamicha's discards instead, only our turn of draw is skipped. In other words, calling shimocha's discards benefits our opponents the least. For hands like (328), we can absolutely call  from shimocha, as well as any other stuff that we'd usually hesitate to call for any other hands. This, of course, is not to say that we cannot call from kamicha; it is a thing to be kept in mind and taken into consideration all the time.

(328) All-last, dora ,  is twice cut


Chi  and pon  with (329) is a must, but also keep in mind that we can also chi  ryanmens. Some people may be wary of the possibility of ending up bad waits if we get rid of good shapes, but even then, the bad waits are 2(8) tiles, not too bad as final waits. On the contrary, the bad waits in (330) are middle tile pairs, so do not chi  ryanmens with (330) unless they are drained up (川村晃裕, 2017, pp. 21–22).

(329) All-last, dora ,


(330) All-last, dora ,


 chi is a must for (331). However, besides those,  is also callable choices. To get two tanyao blocks out of , we can see it as  +  or  + . In the former case,  chi becomes obvious, but what is better than  chi is  pon which leaves us a ryanmen that accepts 5 copies rather than a pair that only accepts 1 copy. If kamicha drops  instead, chi leaves us a  pair that still accepts 2 copies of . In the case of  + , the true bottleneck is  ryanmen which gives us furiten risk when we draw , so  is a must and  is a no-go. Notice that manzu part is a sankanchan, but calling  chi destroys manzu part and leaves us two floating tiles (川村晃裕, 2017, pp. 55–56).

(331) All-last, dora ,


3.2 Yakuhai

In most cases, call yakuhai pon. In any of the following cases, refrain from calling (平澤元氣, 2019, p. 103):

- Our yakuhai pair is the only head and calling it would slow us down (unless our hand is full of bad shapes.)

- Our hand is still cheap and far away after calling

Unlike kuitan, if we skip one yakuhai call, it would be very difficult for the last tile to come out again. To have our yaku confirmed, the criteria for yakuhai pon is generally lower than kuitan.

Yakuhai Pair is The Only Head

For hands like (332) which has only one pair it does not make our hand faster if we pon it. No matter how many times it is cut, do not pon (平澤元氣, 2019, p. 104).

(332) East 1, West seat, dora

For hands such as (333), even though it is close to riichi, its value is 2600 minimum, which is just 600 points higher than open hand. If it is mid round, we must pon ; calling pon in early turns would not hurt too much (平澤元氣, 2019, pp. 105–106).

(333) East-1, West seat, dora

Hand (334) has a bad shape , so getting rid of the only head will also make part of our acceptance. In cases like this where we have bad shapes in our hand, deliberately getting rid of the head actually speeds our hand up.

(334) East-1, West seat, turn 5, dora

The same principle applies to shapes after yakuhai pon. If we pon with (335), our acceptance changes from to . This is thus a must call. If our hand has one ryanmen like (336), or if our hand has an anko like (337), calling pon is still better (平澤元氣, 2019, p. 199).

(335) East-1, West seat, dora

(336) East-1, West seat, dora

(337) East-1, West seat, dora

Hand (338) is already worth 7700 points. We should pon  first to have our yaku confirmed (独歩他, 2017a, pp. 164–167).

(338) East-1, West seat, turn 4, dora 


Cheap and Far Away

If our hand is 3-away, is cheap, has no good shape, and has no pair other than yakuhai, refrain from calling (平澤元気, 2019, p. 110). For hands like (339),  pon is a big no-go.

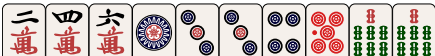
(339) East-1, West seat, turn 3, dora 


Atozuke

Atozuke refers to the state where only a part of our waits give us a yaku. For example, (340) waits on  but only  tsumo or ron allows us to win this hand. In a ruleset called *ariari*⁴², atozuke is allowed and all fine and dandy.

The issue would be how and why we approach atozuke. For example, is there anything we can call other than  in (341)? Do we call any of  from kamicha? Would the number of dora in our hand change how much we want to go for atozuke?

(340)  

(341)  

Some people may shun atozuke, afraid of the unstable state and attention from other opponents. However, know that even if your hand looks like anything but kuitan, your opponents do not always try to strangle you for that. The criteria for atozuke is as follows (独歩他, 2017a, pp. 146–151),

- Always call bad shapes (unless someone calls riichi first) in any of the following situations,
 - Agaritoppu
 - First place in south round
 - Hands with mangan value
- If we are not in the lead and our hand is not mangan either,
 - Call only when it gives us 2000 1-away
 - Call only when it gives us 3900 2-away

For hands like (342), its value does not justify loss in defense after  call except for agaritoppu.

⁴²The first *ari* refers to kuitan, and the second *ari* refers to atozuke. Some rulesets ban both, and those rulesets are called *nashinashi*. By the way, there is no such thing as *arinashi* or *nashiari* which only bans one of kuitan and atozuke.

(342) Turn 4, dora 



For hands like (343) which has 1 dora, opinions may vary from person to person. Since calling  only gives us 2000 2-away and thus does not meet the criteria shown previously, skipping those should be fine.

(343) Turn 4, dora 

The hand consists of 13 tiles: two red dragons, four fours, seven sevens, seven eights, two twos, one pair of fives, one pair of sixes, one pair of sevens, one pair of eights, three nines, and two north winds.

If we have 2 doras like (344), calling any of    is fine.

(344) Turn 4, dora  

If opened parts such as  from (345) would alarm your opponents and make them hesitate to drop live dora  chi would not be wise. If your opponents do not care even a little bit,  chi is fine.

(345) East-1, West seat, turn 4, dora

Do not call  chi from (346). Even though we have dora 3 and we would like to move forward as fast as possible,  chi does not get rid of bad shapes, and it reduces the number of tiles that could give us another pair. Calling any of the kanchan is acceptable (福地誠, 2017, pp. 116–117).

(346) Non-dealer, turn 2, dora 



If we have 3 pairs other than yakuhai pair, we can call any of them. With 4 pairs in our hand, toitoi is also within reach; atozuke with yakuhai even if we do not get to score toitoi is also fine. Call  pon with (347) (ゆうせー, 2020e).

(347) East-2, turn 1, dora 



Situation 2 shows two copies of  already seen. Kamicha drops a tile; it could be  or  (or any other things.) If it is , since  is not critical, we can skip it just fine, and  are likely useless for kamichi who will drop those later. However, if kamicha drops  instead, it is another story. If we choose to skip it, there is only one copy of  left somewhere in the wall, and our  ryanmen becomes almost as weak as a kan . While it is true that a visible    may

alarm others and make  less likely to come out, skipping  also weakens our waits. Likewise, we should call outside tiles such as 2 and 3 more compared to their inside suji counterparts (5 and 6) (川村晃裕, 2017, pp. 131–132).

Situation 2: Call kamicha's , but do not call 

it would also be difficult for other opponents to make use of , making it a good wait (川村晃裕, 2017, pp. 121–123).

(350) East-1, south seat, early turns, dora 

3.4 Toitoiho

If we are chiitōi 1-away and also have a yakuhai pair, generally calling is better. The more 1289 pairs we have in our hand, the better is to call. Therefore, with (351) if someone drops , we must call it and call any other pairs (独歩他, 2017a, pp. 204–211).

(351) East-1, West seat, turn 4, dora 

															
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If we force toitoi with (352), we will not be able to make us of dora  and our hand will only be 2600. Unless we want to confirm a yaku as soon as possible, do not open this hand.

[illegible]

If we are given 4 pairs in the haipai (353) and oya drops  first, calling  pon is fine. If your opponents get alarmed easily, skipping pon is also fine. For some people, 1 yakuhai pair does not qualify as a toitoi hand (福地誠, 2017, pp. 132–133, p 217).

(353) Non-dealer, dora 



Hand (354) has 4 pairs and one dora anko, but 3 of its pairs are middle tiles and hard to call. There are people who wants to call 𠄎 on turn 1, and there are also people who wants chitoi even at turn 6 (福地誠, 2017, pp. 134–135, p. 217).

(354) East round, non-dealer, dora 



3.5 Kan

Any of the following calls are collectively referred to as kan:

- Daiminkan: claiming others discards by opening an anko in our hand.
- Kakan: when we draw the 4th copy of a tile which we already pon'd. This is sometimes referred to as *shouminkan*.

- Ankan: when we draw the 4th copy of a tile which we already have an anko of.

An opened quadruplet is called *minkantsu* or *minkan* for short, and a closed quadruplet is called *ankantsu* or *ankan* for short.

The main benefits of kan is to increase hand value, either through the kan dora that is revealed after the kan call is completed, or through the additional fu from the quadruplet. For example, a terminal anko gives 8 fu, and a terminal ankan gives 32 fu. Another benefit of kan is an additional draw, which is why when we are tenpai with ryanmen kan is usually better. Kan dora also benefits our opponents, and it is three times likely that someone will get kandora than we getting it. Therefore, kan far away from tenpai is not wise and often advised against.

Daiminkan

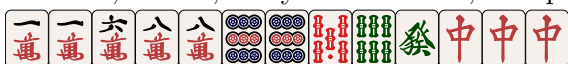
When 3 out of the following 4 criteria is met, we can commit daiminkan (川村晃裕, 2017, p. 140):

1. We are in last place and in dire need of points
2. It is unlikely for us to stay menzen (for example, only one completed block and we'd call everything else)
3. By calling, we can get faster than others
4. At least 2 hans

The reason why 2 hans would be the sweetspot of kan is due to the value brought by the kan dora. Lets say we only have a chun ankou and call daiminkan with it, with 1 kan dora our hand value jumps from 1300 to 2600 (or 2000 to 3900 for dealer.) Considering the value increase of others who likely have doras, this 1300 points of difference is easily outnumbered.

For hands like (355), since we have 1 dora, in last place, and all other pairs are perfect for calls, we can commit 中 daiminkan (川村晃裕, 2017, pp. 139–140).

(355) South-3, dealer, early to mid turn, last place, dora 西



Hand (356) has no dora, but we can force honitsu with the hand to make it at least 3 han, and any kan dora increase our hand value by a lot. Therefore, we can kan 中. However, if any of 西, 東, 南, 北 becomes new dora, we don't need to force honitsu anymore and we still have mangan, in which case just follow pure efficiency (川村晃裕, 2017, pp. 141–142).

(356) Last place, dora 西



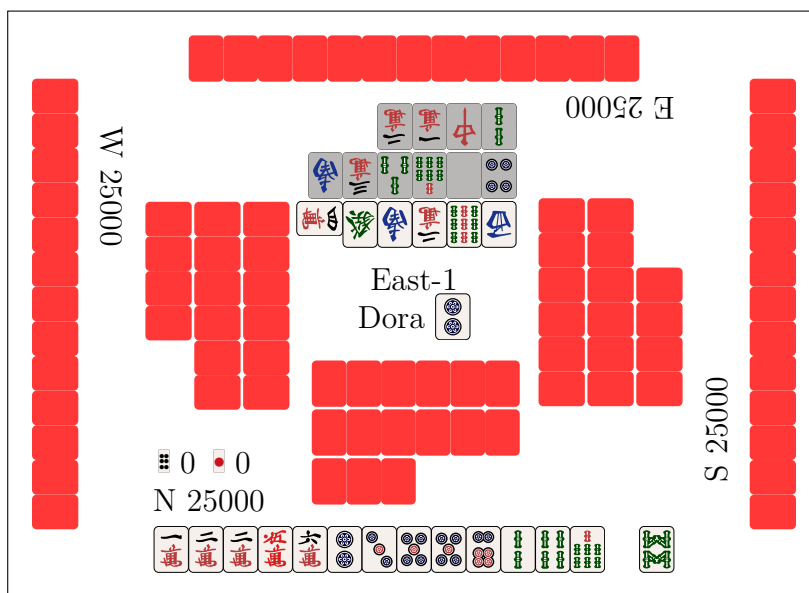
Hand (357) has 2 dora, but our hand is also close to menzen tenpai, so we should not kan 中 unless someone calls or it is mid turn. Even so, only kan shimocha's 中 and skip toimen and kamicha's draws. If we only have one aka dora, the value brought by riichi is bigger than kan, so do not kan at all (川村晃裕, 2017, pp. 141–142).

(357) Last place, dora 

3.6 Canceling Ippatsu and Shifting Haitei

The only effective ippatsu canceling is when someone can overtake us with haneman tsumo (平澤元気, 2019, p. 188). In other words, when we are in first place, have a lot of safe tiles, and second place who is 14000 below us calls riichi, we can go make any calls to cancel their ippatsu. In other cases, deliberately shortening our hand and giving up our tenpai chance would not worth it.

Always shift haitei (平澤元気, 2019, p. 190), especially when our hand is hopeless. However, shifting haitei only makes a difference of 100 points in terms of expected value (みーにん・梶本琢程, 2019, p. 188). In Situation 4 where we have enough genbutsu, if we drop  we won't be able to chi , and dropping any of  makes us unable to call genbutsu from kamicha. Therefore, drop  here. Since 2 copies of  are already on the table, there is no way we can pon it.



Situation 4: When choosing a genbutsu to drop, choose the one that gives us the most chances to call and shift haitei. Drop .

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